RADIATION THERAPY BOARD PREP

Registry Review

A Student's Friendly Guide to the ARRT® Radiation Therapy Certification Exam

First Edition · 2026

All eight content areas · worked examples · self-check questions

A companion study guide to the RT Image Matching Trainer

Contents

How to Use This Book

Welcome. This book is a friendly, plain-language guide to everything the ARRT® Radiation Therapy certification exam asks you to know. It is written for students — people who are still building their mental map of radiation therapy — so it explains ideas from the ground up, uses everyday analogies, and never assumes you already memorized the jargon.

The book is organized to mirror the exam's own structure. The ARRT groups its content into three big areas — **Patient Care**, **Safety**, and **Procedures** — which split into eight subtopics. Each of the eight chapters in this book maps to one of those subtopics, so you always know how what you are reading connects to what you will be tested on.

Here is how to get the most out of each chapter:

- **Start with "What you'll learn."** These objectives tell you what success looks like before you dive in.
- **Read for understanding, not memorization.** The goal is to get the idea. Numbers and lists stick far better once the concept underneath them makes sense.
- **Stop at every "Key Point."** These boxes flag the ideas most worth remembering.
- **Watch for "Common mix-up" boxes.** They head off the specific confusions that trip students up on exam day.
- **Do the "Check yourself" questions.** Cover the italic answer, try the question, then reveal it. Active recall is one of the most powerful study tools there is.
- **Use the references.** Every chapter ends with the authoritative sources behind its facts, so you can dig deeper or confirm anything for yourself.

A quick word on the numbers in this book. In radiation therapy, the exact dose and number of treatments for a given cancer vary from clinic to clinic and from protocol to protocol. When you see a specific dose here, read it as a commonly used example, not the single "correct" answer. The exam — and real practice — care far more that you understand the **principles**.

> **Key Point:** This is an independent study guide. The practice explanations are original and are **not** actual exam questions, and this book is **not affiliated with, endorsed by, or sponsored by the ARRT**. "ARRT" is a registered trademark of the American Registry of Radiologic Technologists.

The Exam at a Glance

The ARRT Radiation Therapy exam is a computer-based, multiple-choice test. It contains **200 scored questions** (plus 30 unscored "pilot" questions the ARRT is trying out, which do not count toward your score), and you get about **3.5 hours**. Scores are reported on a scaled system, and you need a scaled score of **75** to pass.

The 200 scored questions are divided across the three content areas like this:

Content area	Scored questions	Share of exam	Subtopics (this book's chapters)
Patient Care	46	~23%	Patient Interactions & Management (Ch. 1) · Patient & Medical Record Management (Ch. 2)
Safety	51	~25%	Radiation Physics & Radiobiology (Ch. 3) · Radiation Protection, Equipment Operation & QA (Ch. 4)
Procedures	103	~52%	Treatment Sites & Tumors (Ch. 5) · Treatment Volume Localization (Ch. 6) · Prescription & Dose Calculation (Ch. 7) · Treatments (Ch. 8)

Notice that **Procedures is more than half the exam.** If you are budgeting study time, that is where the most questions live — especially treatment delivery, dose calculation, and getting the patient and target lined up correctly. Patient Care and Safety together make up the other half, and they are very learnable: a lot of it is careful, sensible, day-to-day practice.

> **Key Point:** The single best predictor of passing is steady, understanding-first study across **all** three areas — not cramming the math. Every chapter here is sized to be read in one or two sittings.

One more note: the ARRT periodically updates its content outline. A revision takes effect in early 2027 that keeps the same three-area structure and adds a few modern topics (3D/4D simulation, the electronic medical record, bone-metastasis palliation, mental-health-crisis communication, and oxygen administration, among others). Those topics are woven into the relevant chapters here so you are covered either way.

Chapter 1 — Patient Interactions & Management

What you'll learn

- How to communicate clearly with patients of every age, background, and reading level, and how to read the emotional cues that words don't say.
- Your role in informed consent, psychosocial support, and crisis escalation — what you do, and what belongs to the physician.
- How to measure and interpret vital signs, recognize acute symptoms and radiation side effects, and respond to medical emergencies.
- Safe, dignified patient transfers and immobilization, including managing lines, tubes, and oxygen.
- The essentials of iodinated contrast media: types, reactions, and pre-screening.
- The ethics, law, and infection-control rules that frame every patient encounter.

Radiation therapy is a technical profession, but most of your day is spent with people — often frightened, tired, or in pain. The treatment plan only works if the patient shows up, holds still, and trusts you enough to come back tomorrow. Mastering patient interaction is not "soft skills"; it is the foundation on which accurate, safe treatment is built [1].

Communication: meeting the patient where they are

A **radiation therapist** (the credentialed professional who delivers daily radiation treatments and operates the treatment machine) talks to patients constantly. The goal is always understanding, not just speaking.

Health literacy

Health literacy is a person's ability to obtain, process, and understand basic health information well enough to make decisions. Many patients have low health literacy and will never tell you so. The safest approach is to use plain words by default: say "the area we are treating" instead of "the treatment field," "feeling sick to your stomach" instead of "nausea."

A reliable check is the **teach-back method**: instead of asking "Do you understand?" (to which almost everyone says yes), ask the patient to explain it back in their own words — "Just so I know I explained it well, can you tell me what you'll do if your skin gets sore?" If they can teach it back, they've got it.

> **Key Point:** Match your words to the listener, not to your textbook. Default to plain language and confirm with teach-back.

Active listening and emotional cues

Active listening means giving your full attention, reflecting back what you hear, and noticing the feelings underneath the words. Watch for **nonverbal cues** — body language, tone, and facial expression. A patient who says "I'm fine" while gripping the table and avoiding eye contact is not fine. Name the emotion gently: "This part can feel a little overwhelming. How are you doing?" Naming a feeling out loud often relieves it.

> **Common mix-up:** Active listening is not the same as staying silent. Silence alone can feel like you've checked out. Active listening includes brief, genuine responses — nodding, summarizing, asking a follow-up — that show you heard.

Reviewing pertinent medical history

Before treatment you'll review the **pertinent medical history** — the parts of the patient's record that affect today's care: allergies (especially to contrast or latex), current medications, implanted devices such as pacemakers, pregnancy status, recent surgeries, and mobility limitations. Confirm identity using **two patient identifiers** (for example, full name and date of birth) every single time. Never rely on the patient simply answering "yes" when you call a name — ask them to state it.

Coaching CT-simulation compliance

CT simulation ("CT sim") is the planning session where the patient is positioned and scanned to design the treatment. The position chosen here is the one they must hold for weeks, so coaching matters.

Explain what will happen before it happens: the table will move, the scanner is open (not a closed tunnel), small marks or tiny permanent **tattoos** may be placed on the skin for alignment, and they must hold still. Tell them how long, and give them a signal to use if they need help. Patients comply best when they know what to expect and feel in control.

Patient education and informed consent

Informed consent is the process by which a patient agrees to treatment after being told its purpose, benefits, risks, and alternatives [6].

Here is the rule that shows up again and again:

> **Key Point:** The **physician obtains informed consent.** The therapist reinforces the explanation, verifies the patient's understanding, and documents it. You never sign the consent for a procedure on the physician's behalf.

If a patient says "Wait — what are the risks again?" the right move is not to wing it. Answer what you can within your scope, and route remaining questions back to the physician before treatment proceeds. Your patient-education role is real and important; it simply sits beside the physician's legal duty, not in place of it [1].

Psychosocial support and crisis escalation

A cancer diagnosis carries enormous emotional weight. **Psychosocial** refers to the combined psychological and social dimensions of a person's life. **Distress** is the expected emotional response — sadness, fear, anxiety — and most patients feel it.

Your job is to support and to refer. You are not the patient's therapist, but you are often the person they see most. When distress exceeds normal coping, connect them with social work, chaplaincy, or psycho-oncology services per your department's pathway.

> **Key Point:** Any sign of a **mental-health crisis** — talk of suicide, self-harm, or harming others — is escalated immediately and never left unattended. Stay with the patient, notify the physician/nurse, and follow your facility's emergency protocol.

Cultural competence and interpreter use

Cultural competence is the ability to deliver care that respects each patient's beliefs, values, and language. Some patients have customs around modesty, gender of caregivers, diet, or family decision-making that affect care — ask respectfully rather than assume.

When there is a language barrier, use a **qualified medical interpreter** (in person or via phone/video). Do not use a family member, and especially not a child, as the interpreter for medical information: they may filter, soften, or misunderstand. Speak to the patient, not the interpreter ("How are you feeling today?" not "Ask her how she's feeling").

Age-specific care

Communication and risk shift across the lifespan.

Pediatric patients

Children need **developmentally appropriate** communication — language and explanations matched to their age and understanding. Use simple words, show rather than tell, and let a **child-life specialist** (a professional trained to help children cope with medical care through play and preparation) assist when available. Some young children need anesthesia or sedation to hold still.

A child cannot legally consent. A **surrogate decision-maker** — usually a parent or legal guardian — consents on the child's behalf. Still involve the child at their level so they feel respected, not done-to.

Geriatric patients

Older adults may face **frailty** — reduced physiologic reserve that raises the risk of complications. Screening tools flag who needs extra support. Common ones include the **G8**, the **VES-13** (Vulnerable Elders Survey), and the **Edmonton Frail Scale**. Also assess **fall and mobility risk** before any transfer: gait, balance, vision, and medications all contribute.

Population	Watch for	Helpful resource
Pediatric	Fear, inability to hold still, need for sedation	Child-life specialist; caregiver as surrogate
Geriatric	Frailty, falls, sensory/cognitive decline	G8 / VES-13 / Edmonton screening

Vital signs and normal ranges

Vital signs are basic measurements of body function. Memorize typical adult resting ranges and remember that "normal" varies with age, fitness, and condition.

Vital sign	Typical adult resting range
Temperature	~36.5–37.5 °C (97.7–99.5 °F)
Heart rate (pulse)	60–100 beats/min
Respiratory rate	12–20 breaths/min
Blood pressure	~90/60 to 120/80 mmHg
Oxygen saturation (SpO_2)	95–100%

Common mix-up: Bradycardia vs. tachycardia. **Brady-** means slow (heart rate under ~60), **tachy-** means fast (over ~100). The "brady" in "brady-cardia" can anchor in your memory as "brakes on."

Acute symptom management

You will recognize and respond to common acute symptoms. Know the basics; defer medication decisions to the physician/nurse.

- **Nausea and vomiting:** keep the patient safe from aspiration (head turned, suction available), offer a basin, notify the team; antiemetics are physician-ordered.
- **Fatigue:** common and cumulative during a course; encourage rest, light activity, and reporting of worsening tiredness.
- **Dyspnea (shortness of breath):** sit the patient upright, stay calm, check SpO_2, and call for help if it persists or worsens.

- **Pain:** assess on a 0–10 scale, report uncontrolled pain, and position for comfort.
- **Syncope (fainting):** if a patient feels faint, lower them safely to prevent a fall, lay them flat with legs elevated if appropriate, and call for help.

Acute vs. late radiation side effects

Distinguishing these is a high-yield concept.

- **Acute side effects** appear during or shortly after treatment, in rapidly dividing tissues. Examples: skin reactions (erythema, moist desquamation), mucositis, fatigue, nausea. They usually heal after treatment ends.
- **Late side effects** appear months to years later in slower-responding tissues. Examples: fibrosis (scarring), telangiectasia (tiny dilated vessels), strictures, and secondary malignancy. They are often permanent.

> **Common mix-up:** Acute ≠ severe. "Acute" describes timing (early onset), not how serious the effect is. A late effect can be far more serious than an acute one.

Comfort, dignity, and modesty

Every interaction protects the patient's dignity. Knock before entering, explain what you're doing and why, expose only the area needed, and drape the rest. Provide warmth and privacy. These small acts build the trust that keeps patients returning for a full course of treatment.

Patient transfer and body mechanics

Moving patients safely protects both of you.

- An **assisted transfer** is for a patient who can bear some weight and help; you guide and steady them.
- A **dependent transfer** is for a patient who cannot help; use a lift device, a slide board, or enough trained staff — never your back.

Use proper **body mechanics**: keep a wide base of support, bend at the knees (not the waist), keep the load close to your body, and pivot your feet instead of twisting your spine.

> **Key Point:** When in doubt, get help and use a mechanical lift. No transfer is worth a patient fall or a career-ending back injury.

Managing lines, tubes, and oxygen during moves

Before any move, account for every attachment: IV lines, urinary catheters, chest tubes, drains, and oxygen tubing. Give them slack, keep drainage bags below the insertion site, never let a line stretch or kink, and confirm oxygen flow is uninterrupted. A dislodged line during a transfer is a preventable emergency.

Immobilization devices and breath-hold

Immobilization keeps the patient in the same position every day so the beam hits the target accurately.

- **Thermoplastic mask:** a plastic mesh, softened in warm water, molded to the patient (often head/neck), then hardened to hold them still. Warn patients it can feel snug or warm; coach claustrophobic patients carefully.
- **Breast board:** an angled board that positions the arms overhead and the chest at a set incline for chest/breast treatments.
- **Vacuum bag (vac-bag):** a cushion filled with tiny beads; air is suctioned out so it molds rigidly around the body.

Breath-hold techniques — such as **deep-inspiration breath-hold (DIBH)** — ask the patient to take in a breath and hold it, moving the target (for example, the heart away from a left breast field). Success depends entirely on clear, calm coaching: practice with them, give consistent cues, and reassure them they control when to release.

Medical emergencies and BLS/ACLS readiness

You must recognize emergencies and act while help is summoned.

- **Syncope:** protect from falling, position flat with legs raised, check responsiveness and breathing.
- **Respiratory distress:** sit upright, administer oxygen per protocol/order, call for help.
- **Oxygen administration:** common low-flow delivery is a **nasal cannula** (typically 1–6 L/min); a **non-rebreather mask** delivers a high concentration for emergencies. Keep oxygen away from flame and grease.
- **Chest pain:** treat as cardiac until proven otherwise — stop, call for help, keep the patient still and calm, get the emergency team.
- **Contrast reactions:** see below.

Maintain current **BLS** (Basic Life Support — CPR and AED skills) certification; many departments also expect familiarity with **ACLS** (Advanced Cardiac Life Support) team roles [7]. Know where your crash cart, AED, oxygen, and emergency call button are before you need them.

Contrast media

Contrast media are agents that make structures more visible on imaging. In radiation therapy you'll most often encounter **iodinated contrast** used in CT.

- **Ionic** agents have higher osmolality and a somewhat higher reaction rate; **non-ionic** agents are lower-osmolality and better tolerated. Most modern CT uses non-ionic contrast.

Reactions range in severity:

Severity	Examples	Response
Minor (mild)	Warmth, flushing, mild hives, nausea	Observe, reassure; usually self-limited
Moderate	Widespread hives, vomiting, mild wheezing, tachycardia	Notify physician, treat symptoms, monitor closely
Severe	Laryngeal edema, severe bronchospasm, anaphylaxis, cardiovascular collapse	Emergency: call for help, support airway/circulation, epinephrine per physician

Pre-screening before iodinated contrast checks: **renal function** (kidney function, often via eGFR/creatinine, because the kidneys clear the agent), prior **allergy** or contrast reaction, and **pregnancy** status. A documented prior severe reaction is a major red flag the physician must address before any further contrast.

> **Common mix-up:** A shellfish allergy does not specifically predict iodinated-contrast reaction. The old "iodine allergy" idea is outdated; what matters most is a prior reaction to contrast itself and a general atopic/allergic history.

Ethics and law

Several frameworks govern every encounter.

- **Patient Bill of Rights:** patients have rights to respectful care, information, privacy, and participation in decisions.
- **Autonomy and the right to refuse:** a competent adult may **refuse** any treatment, even after starting. Honor it, notify the physician, and document; never coerce.
- **HIPAA** (Health Insurance Portability and Accountability Act): the **Privacy Rule** protects how health information is used and shared, and the **Security Rule** protects electronic health information [3]. Share patient information only on a need-to-know basis. Don't discuss patients in hallways or elevators.
- **ARRT Standards of Ethics:** the profession's Code of Ethics (aspirational) and Rules of Ethics (enforceable) that guide honest, competent, patient-centered practice [2].

- **ASRT Scope of Practice / Practice Standards:** define what a radiation therapist is qualified and expected to do — stay within it, and refer beyond it [1].

> **Key Point:** Autonomy means the patient — not you, and not the family — decides. A competent adult's refusal is final once informed.

Infection control

Preventing infection protects vulnerable, often immunocompromised patients [4].

- **Standard Precautions:** treat all blood and body fluids as potentially infectious for every patient. This includes **hand hygiene** (the single most effective measure) and appropriate **PPE** (personal protective equipment — gloves, gown, mask, eye protection as needed).
- **Hand hygiene:** wash with soap and water when hands are visibly soiled or after caring for certain infections (e.g., C. difficile); alcohol-based rub is fine otherwise.
- **Transmission-based precautions** add protection for known pathogens:

Type	Use for (examples)	Key PPE/measure
Contact	MRSA, C. diff	Gloves + gown; dedicated equipment
Droplet	Influenza, pertussis	Surgical mask within ~3–6 ft
Airborne	TB, measles, varicella	N95 respirator; negative-pressure room

- **Neutropenic precautions:** for patients with very low white-blood-cell counts (common after chemo). The goal is to protect the patient from you and the environment — meticulous hand hygiene, no sick visitors, no fresh flowers, and careful cleaning.

> **Common mix-up:** Neutropenic ("protective") precautions vs. isolation precautions. Isolation keeps an infection in (protecting others); neutropenic precautions keep infection out (protecting the fragile patient). Same diligence, opposite direction.

Check yourself

1. Who obtains informed consent, and what is the therapist's role? The physician obtains informed consent. The therapist reinforces the explanation, verifies the patient's understanding (e.g., with teach-back), and documents it.

2. A patient says "I understand" but you're not sure. What technique confirms it? The teach-back method — ask the patient to explain the information back in their own words.

3. Define acute vs. late radiation side effects and give one example of each. Acute effects occur during/shortly after treatment in fast-dividing tissue (e.g., skin erythema, mucositis) and usually heal. Late effects appear months to years later in slow-responding tissue (e.g., fibrosis, telangiectasia) and are often permanent.

4. List the three core items to pre-screen before iodinated contrast. Renal (kidney) function, prior allergy/contrast reaction history, and pregnancy status.

5. Why should you use a qualified medical interpreter instead of a family member? A family member (especially a child) may filter, soften, or misunderstand medical information; a qualified interpreter conveys it accurately and impartially, and it protects privacy.

6. What direction of protection do neutropenic precautions provide? They protect the immunocompromised patient FROM environmental and caregiver pathogens — keeping infection out — rather than containing a patient's infection.

Chapter references

Free, full-text sources

1. ASRT — Practice Standards for Medical Imaging & Radiation Therapy: https://www.asrt.org/main/standards-and-regulations/professional-practice/practice-standards-online (scope of practice, patient care).
2. ARRT — Standards of Ethics: https://www.arrt.org/pages/arrt-reference-documents/by-document-type/standards-of-ethics.
3. U.S. HHS — HIPAA for Professionals (Privacy & Security Rules): https://www.hhs.gov/hipaa/for-professionals/.
4. CDC — Infection control in outpatient/oncology settings: https://www.cdc.gov/infection-control/.
5. American Heart Association — CPR & ECC (BLS): https://cpr.heart.org/.
6. ACR–ASTRO Practice Parameter on Informed Consent in Radiation Oncology (open access): https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10425961/.

For deeper reading (library / textbook)

1. Washington CM, Leaver DT. Principles and Practice of Radiation Therapy. Elsevier — patient care, immobilization, and clinical workflow chapters.

Chapter 2 — Patient & Medical Record Management

What you'll learn

- How cancer risk factors, common cancer sites, and TNM staging shape the treatment plan you'll deliver.
- How to review a medical history and read a pathology report without getting lost in the jargon.
- Which imaging studies and lab values planners rely on, and what they each tell you.
- What a complete radiation prescription must contain, and why the daily treatment record is a legal document.
- How independent monitor-unit checks, physics chart checks, and incident reporting keep patients safe.
- Why the radiation therapist is the team's daily eyes and ears on the patient.

Why this matters

Before a single beam turns on, an enormous amount of information has already been gathered, checked, and written down. As the therapist, you are the last human safeguard between that record and the patient on the table. Understanding where each piece of information comes from — and how to spot when something doesn't add up — protects your patients and is heavily tested on the ARRT Radiation Therapy exam.

Cancer Epidemiology and Risk Factors

Epidemiology is simply the study of how often a disease occurs and who it affects. Knowing the patterns helps you understand why certain patients sit in your waiting room.

Risk factors fall into a few broad groups:

- **Lifestyle:** Tobacco is the single largest preventable cause of cancer, linked to lung, head and neck, bladder, and many other sites. Alcohol (especially combined with tobacco), obesity, poor diet, and physical inactivity also raise risk. Ultraviolet light from the sun drives skin cancers.
- **Occupational and environmental:** Asbestos (mesothelioma and lung), radon gas, benzene, and certain industrial chemicals. Ionizing radiation exposure itself is a risk factor — which is exactly why we treat radiation with such respect.
- **Infectious agents:** Some viruses cause cancer. The big three to memorize are **HPV** (human papillomavirus), the leading cause of cervical and many oropharyngeal — back of the throat — cancers; **HBV and HCV** (hepatitis B and C viruses), which drive liver

cancer; plus H. pylori bacteria (stomach) and Epstein-Barr virus (certain lymphomas and nasopharyngeal cancer).

> **Key Point:** HPV-positive oropharyngeal cancers generally respond better to treatment and carry a more favorable prognosis than HPV-negative ones. That single status can change the whole plan.

Common cancer sites

In the United States, the cancers you'll see most often are **breast, lung, prostate, and colorectal**. Skin cancer is the most common of all but is frequently treated outside the radiation oncology department. Knowing the common sites helps you anticipate the imaging, positioning, and side effects typical of each.

Why TNM Staging Matters

Staging describes how far a cancer has spread. The international standard is the **TNM system**, maintained in the AJCC Cancer Staging Manual [2]:

Letter	Stands for	Asks
T	Tumor	How large is the primary tumor / how deep has it invaded?
N	Nodes	Are regional lymph nodes involved, and how many?
M	Metastasis	Has it spread to distant organs?

These combine into an overall **stage I through IV**, where higher means more advanced. Think of TNM as a shared language: a "T2 N1 M0" tumor means the same thing in New York as it does in Tokyo. Staging drives the **intent** of treatment (cure versus comfort), the dose, and which fields are treated.

> **Key Point:** Staging is determined before treatment and does not change even if the tumor shrinks. A restaged tumor gets a new prefix (like "yp" for post-therapy pathologic stage), not a rewritten original stage.

Reviewing the Medical History

A thorough history protects the patient. When you review a chart, look for:

- **Prior cancer treatment.** Previous **surgery**, **chemotherapy** (drug therapy), or **radiation therapy (RT)** all matter. For prior RT, you need the **site, dose, and dates** — re-irradiating tissue that already received its lifetime limit can cause serious harm.
- **Comorbidities.** Other medical conditions such as diabetes, heart disease, or autoimmune disorders that affect tolerance to treatment.

- **Medications and allergies.** Including contrast dye allergies, which matter before a CT simulation with contrast.
- **Pregnancy and lactation.** Always confirm pregnancy status in patients who can become pregnant before any imaging or treatment. Radiation to a fetus can cause severe harm.
- **Performance status.** A simple score of how well a patient functions day to day. Two common scales are **ECOG** (0 = fully active, 4 = completely disabled) and **Karnofsky** (100 = normal, 0 = death). It helps the team judge whether a patient can tolerate aggressive treatment.

> **Common mix-up:** ECOG runs low-to-high as patients get sicker (0 is best), while Karnofsky runs high-to-low (100 is best). They move in opposite directions.

Reading a Pathology Report

The **pathology report** is the diagnosis in writing, produced after a pathologist examines tissue under a microscope. Key terms:

- **Histology:** The cell type and tissue of origin (for example, adenocarcinoma or squamous cell carcinoma).
- **Grade:** How abnormal the cells look. Low grade resembles normal tissue and grows slowly; high grade looks wild and grows fast. Grade is about appearance; stage is about spread.
- **Margins:** The edge of surgically removed tissue. **Negative (clear) margins** mean no tumor at the edge; **positive margins** mean tumor reaches the cut edge and often calls for more treatment.
- **Nodal status:** How many lymph nodes were examined and how many contained cancer.
- **Lymphovascular invasion (LVI)** and **perineural invasion (PNI):** Tumor found inside small vessels or wrapped around nerves. Both suggest a more aggressive cancer and may push toward larger fields or higher dose.

Breast biomarkers: ER, PR, HER2

For breast cancer, the report lists **receptor status**:

- **ER and PR** (estrogen and progesterone receptors): If positive, the tumor may respond to hormone-blocking therapy.
- **HER2** (a growth-signaling protein): If over-expressed, targeted drugs like trastuzumab can be used.

A "triple-negative" breast cancer (ER−, PR−, HER2−) lacks all three targets and is treated more aggressively.

> **Common mix-up:** Grade and stage are not the same. A small, early-stage tumor can still be high grade, and a large tumor can be low grade.

Imaging Used in Planning

Different scans answer different questions:

Modality	Best at	Planning role
CT	Bone, electron density	The backbone of planning — gives the **electron density** map the computer needs to **calculate dose**
MRI	Soft-tissue detail	Defining tumor borders in brain, prostate, and other soft-tissue sites
PET	Metabolic activity	Staging and finding active disease, often fused with CT

CT (computed tomography) is essential because dose calculation depends on how dense each tissue is to the beam, and CT numbers translate directly into that information. **MRI (magnetic resonance imaging)** shows soft tissue beautifully but does not provide electron density, so it's usually fused onto the planning CT. **PET (positron emission tomography)** highlights metabolically active tissue, useful for staging and target definition.

Simulation is the planning session where the patient's position and reference marks are captured. **3D simulation** captures a single static CT; **4D simulation** adds the time dimension, recording how the tumor moves with breathing — crucial for lung and upper-abdominal targets.

Interpreting Labs

You don't diagnose from labs, but you should recognize values that affect treatment safety.

- **CBC (complete blood count):** Checks red cells (low = **anemia**), white cells (low = **infection risk**, since chemo and radiation can suppress them), and **platelets** (low = **bleeding risk**). Very low counts may pause treatment.
- **Liver function tests (LFTs):** Reflect liver health, important before liver or upper-abdomen treatment.
- **Renal panel — BUN, creatinine, and electrolytes:** Kidney function and salt balance. Matters before contrast and for drugs cleared by the kidneys.

- **Tumor markers:** Proteins that can track certain cancers — **PSA** (prostate), **CEA** (colorectal), **CA-19-9** (pancreatic), and **AFP** (liver and germ-cell tumors). They monitor response but rarely diagnose alone.

> **Key Point:** A critically low platelet or white-cell count is the kind of finding you escalate to the physician before treating, not after.

The Radiation Prescription

The **prescription** is the physician's signed order for treatment. A complete prescription must contain:

- **Total dose** and **dose per fraction** (a fraction is one daily treatment), measured in **gray (Gy)**.
- **Fractionation** — the number of treatments and the schedule.
- **Intent:** curative (aiming to cure), adjuvant (added after surgery to mop up), or palliative (relieving symptoms).
- **Modality and energy** — for example, photons or electrons at a stated energy.
- **Organ-at-risk (OAR) constraints** — dose limits for nearby healthy structures.
- **Physician authorization** — a dated signature. Without it, treatment cannot proceed.

Dose and fractionation vary widely by site and protocol. A commonly used breast regimen runs in the neighborhood of 40–50 Gy over several weeks, while a single 8 Gy fraction is commonly used for painful bone metastases. Always follow the **prescribed** values and current clinical guidelines such as NCCN [3]; never treat from memory of "typical" numbers.

> **Key Point:** No prescription, no treatment. An unsigned or incomplete prescription is a hard stop.

Daily Treatment-Delivery Records

Every fraction is documented. The daily **treatment record** is a legal and clinical document that must capture:

- **Date** and **fraction number** (for example, "12 of 25").
- **Monitor units (MU)** delivered — the machine's internal unit of beam output.
- **Daily dose** and running **cumulative dose**.
- **Beam parameters** — energy, field sizes, gantry and collimator angles, accessories.
- **Interruptions** — any pause or missed fraction.
- **Therapist verification** — the initials or signature confirming who delivered the fraction.

This record lets the next therapist, the physicist, and the physician see exactly what the patient has received to date. If it isn't documented, for practical and legal purposes it didn't happen.

Independent MU / Dose Verification

Before treatment begins, the planned **monitor units** are checked by a second, independent method — either separate software or a hand calculation — to confirm the computer's plan is reasonable.

The accepted yardstick comes from ICRU recommendations: the delivered dose should agree with the prescribed dose to within about **±5%**, with **3% or better preferred** [1][6]. If your independent check disagrees with the treatment-planning system by more than the institution's tolerance (often a few percent), you investigate before treating — you never simply override it.

> **Key Point:** The independent MU check exists to catch a planning error before it reaches the patient. A disagreement is a question to answer, not a number to ignore.

EMRs, Physics Chart Checks, and Record Handling

Modern departments run on **electronic medical records (EMRs)** and record-and-verify systems that pull beam parameters directly to the machine, reducing manual-entry errors. With that power comes responsibility: log in as yourself, never share credentials, and protect patient privacy under HIPAA.

The **physics chart check** is a formal review by a medical physicist — verifying the prescription, plan, MU calculation, and documentation — performed at the start of treatment and at regular intervals (commonly weekly). It's a planned, independent double-check of the whole chart.

Incident and Near-Miss Reporting

When something goes wrong — or almost goes wrong — it gets reported. An **incident** is an error that reached the patient; a **near-miss** is one caught before it did. Reporting both, without blame, lets departments fix the underlying process so the same error can't happen twice.

A culture that punishes honest reporting drives errors into hiding. The ASRT Practice Standards [4] frame this as a professional duty: report promptly, document factually, and focus on system improvement.

Common mix-up: A near-miss isn't a non-event to shrug off. It's a free warning — the most valuable kind of report, because no one was harmed yet.

The Therapist as Daily Patient Liaison

You see the patient more than anyone else on the team — often every weekday for weeks. That makes you the first to notice a worsening skin reaction, new pain, weight loss, confusion, or emotional distress. The patient will tell you things they never mention to the physician.

Communicating critical findings is part of the job. A sudden new symptom, a skipped fraction, a setup that won't reproduce, or a lab value flagged critical all need to reach the right team member quickly and be documented. Closing that loop — speaking up and writing it down — is how the chain of safety holds.

Key Point: You are not just running the machine. You are the team's daily eyes and ears, and your observations are part of the medical record.

Check yourself

1. Which three viral infections are most strongly linked to cancers a therapist commonly sees, and to which sites? HPV (cervical and oropharyngeal), and HBV and HCV (liver). H. pylori (stomach) and EBV are also worth knowing.

2. A tumor is staged "T2 N1 M0." What does each letter tell you? T2 describes the primary tumor's size or depth, N1 means regional lymph nodes are involved, and M0 means no distant metastasis.

3. Why is CT the backbone of treatment planning even when MRI shows the tumor better? CT provides the electron-density information the computer needs to calculate dose. MRI lacks that, so it is fused onto the planning CT for soft-tissue detail.

4. List four items a complete radiation prescription must contain. Any four of: total dose, dose per fraction, fractionation schedule, treatment intent, modality and energy, organ-at-risk constraints, and the physician's authorizing signature.

5. Your independent MU check differs from the treatment-planning system by more than your department's tolerance. What do you do? Stop and investigate before treating. ICRU recommends agreement within roughly ±5% (3% or better preferred); a larger discrepancy must be resolved, not overridden.

6. What's the difference between an incident and a near-miss, and why report a near-miss? An incident reached the patient; a near-miss was caught first. You report near-misses because they reveal a system weakness while no one has been harmed — a free chance to fix it.

Chapter references

Free, full-text sources

1. NCI — Cancer Information / PDQ summaries (epidemiology, risk factors, staging, imaging, treatment by site): https://www.cancer.gov/types.
2. ASRT — Practice Standards for Medical Imaging & Radiation Therapy: https://www.asrt.org/main/standards-and-regulations/professional-practice/practice-standards-online.
3. IAEA — *Radiation Oncology Physics: A Handbook for Teachers and Students* (free PDF; treatment-record and dose-reporting context): https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1196_web.pdf.
4. AAPM Reports — monitor-unit verification (e.g., TG-71, TG-114), free: https://www.aapm.org/pubs/reports/.

For deeper reading (library / textbook)

1. AJCC *Cancer Staging Manual*, 8th ed. — the TNM staging reference.
2. Washington CM, Leaver DT. *Principles and Practice of Radiation Therapy*. Elsevier.

Chapter 3 — Radiation Physics & Radiobiology

What you'll learn

- Where therapy radiation comes from — radioactive decay and machine-made x-rays.
- How a beam weakens with distance (the inverse-square law) and with shielding (attenuation and the half-value layer).
- The three ways photons interact with matter, and why **Compton scatter** is the one that matters most in treatment.
- The units we measure radiation in — gray, sievert, and becquerel — and what each one really means.
- How radiation damages cells, why we split treatment into many small fractions, and the math behind it (the linear-quadratic model, BED, and EQD2).
- The difference between deterministic and stochastic effects, and what normal-tissue tolerance means.

Physics is the part of the registry that worries students the most, so we are going to slow down, build each idea from the ground up, and lean on pictures. Nothing here requires fancy math — just a willingness to take it one step at a time. Every concept ties back to a real job you will do at the machine: choosing an energy, keeping yourself safe, and understanding why the prescription is written the way it is.

Part 1 · Where radiation comes from

The atom, in sixty seconds

Everything starts with the atom: a tiny, dense **nucleus** (protons and neutrons, positively charged) surrounded by **electrons** in shells. Radiation therapy is all about adding enough energy to knock electrons loose. When a photon or particle has enough energy to eject an electron from an atom, we call it **ionizing radiation**, and the atom left behind is now an **ion**. That single act — ionization — is the seed of everything that follows, from the image on your screen to the DNA break inside a tumor cell.

> **Key Point:** "Ionizing" means "able to knock electrons off atoms." That is the dividing line between the radiation we use to treat cancer and the harmless radiation of, say, a radio wave.

Radioactive sources and decay

Some atoms are unstable and shed energy over time — they are **radioactive**. The classic therapy example is **Cobalt-60**. Cobalt-60 first emits a beta particle, turning into an excited nickel atom, which then releases its extra energy as **two gamma rays** of 1.17 MeV and 1.33 MeV. (An MeV, mega-electron-volt, is just a unit of energy.) Because the two gammas come out together, we usually treat the average therapy energy as about **1.25 MeV**.

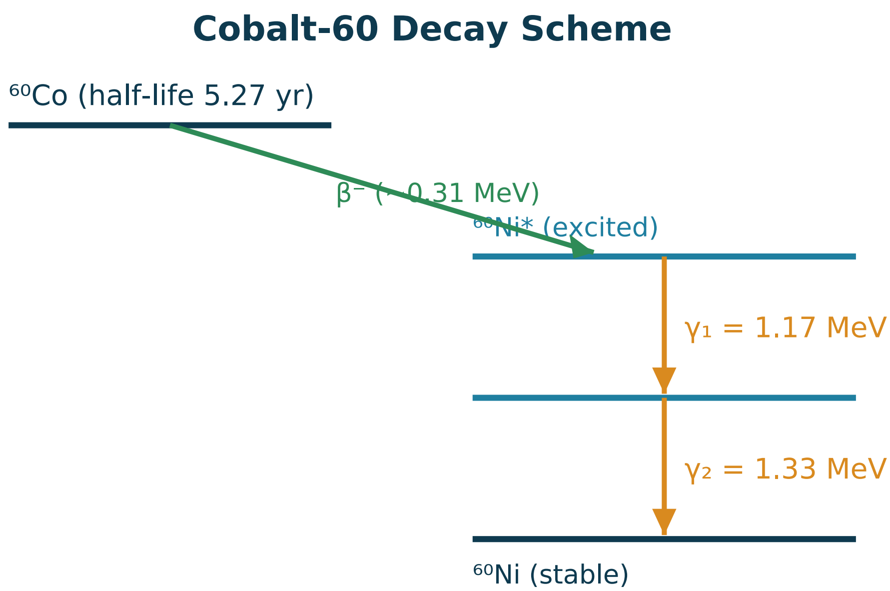

Figure 3.1 — Cobalt-60 decays by beta emission to excited nickel-60, which drops to its stable ground state by releasing two gamma rays. Their average energy (~1.25 MeV) is the effective treatment energy.

Two terms describe a radioactive source:

- **Half-life** — the time for half of the atoms to decay. Cobalt-60's half-life is **5.27 years**, which is why cobalt units must eventually have their source replaced: after about five years, the source is only half as "bright."
- **Activity** — how many atoms decay per second. The modern unit is the **becquerel (Bq)** = 1 decay per second. The older unit is the **curie (Ci)**, where 1 Ci = 3.7×10^{10} Bq.

> **Common mix-up:** Half-life is about time (how fast a source weakens); activity is about rate (how many decays happen each second). They are related but not the same number.

Machine-made x-rays

Most modern treatment uses a **linear accelerator (linac)**, which makes x-rays on demand rather than relying on a radioactive source. It speeds electrons up to high energy and slams them into a metal **target** (usually tungsten). Two things happen:

- **Bremsstrahlung** ("braking radiation," Figure 3.2). An electron passes close to a nucleus, gets deflected and slowed, and the energy it loses flies off as an x-ray. This is the **main** source of the treatment beam. The higher the electron energy, the more — and more penetrating — the x-rays.
- **Characteristic x-rays.** An incoming electron knocks out an inner-shell electron; when an outer electron drops in to fill the gap, it releases an x-ray whose energy is characteristic of that element. These make up a small part of the beam.

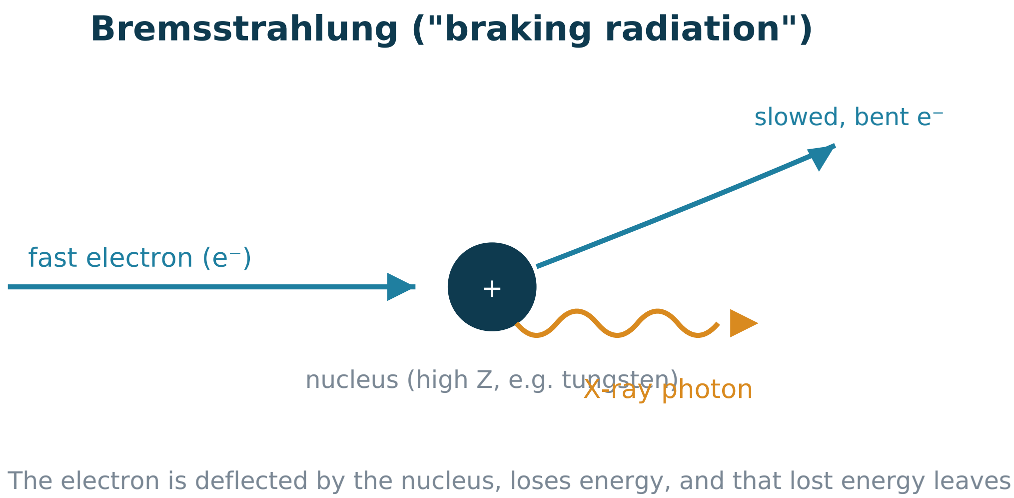

Figure 3.2 — Bremsstrahlung: a fast electron is deflected by a nucleus, loses energy, and that energy leaves as an x-ray photon. This is how a linac builds its treatment beam.

Part 2 · How a beam travels and weakens

Photon energy: kV versus MV

Beam energy is described by voltage. **Kilovoltage (kV)** beams (tens to a few hundred kV) are low energy — good for superficial skin lesions and for the crisp, bony-contrast images you match at the machine. **Megavoltage (MV)** beams (typically 6–18 MV from a linac) are high energy and penetrating — that is what reaches deep tumors. Keeping this kV-versus-MV distinction in mind explains a lot, including why your setup images and your treatment beam look so different.

The inverse-square law

Radiation spreads out from its source like light from a bare bulb. As it travels, the same energy is smeared over a larger and larger area, so the intensity drops — and it drops with the **square** of the distance.

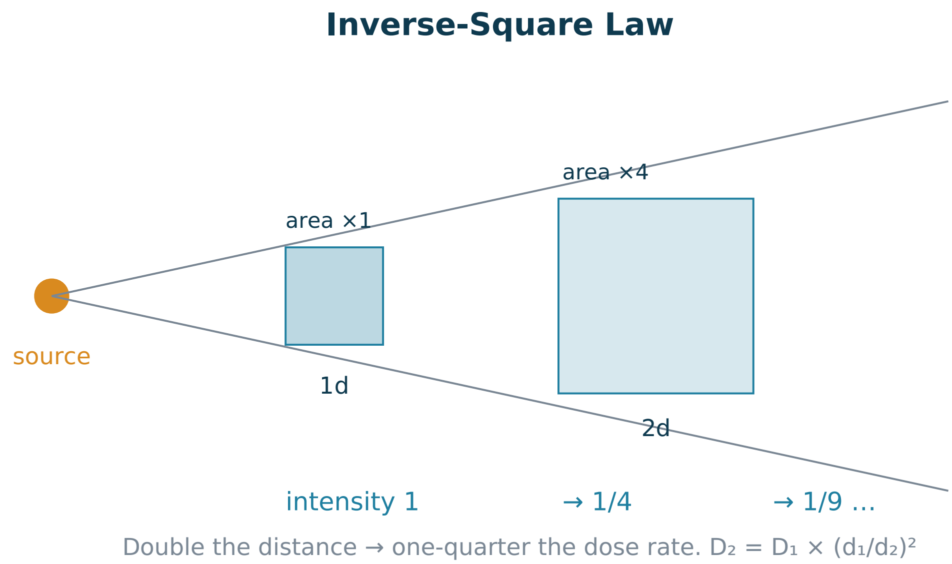

Figure 3.3 — The same rays cover four times the area at twice the distance, so the dose rate falls to one-quarter. At three times the distance it is one-ninth.

We write it as:

> **Key Point: $D_2 = D_1 \times (d_1/d_2)^2$**, where d_1 and d_2 are distances from the source.

Worked example. A point source delivers 100 cGy at 80 cm. What is the dose rate at 100 cm?

```
D2 = D1 × (d1 / d2)²
D2 = 100 × (80 / 100)²
D2 = 100 × (0.8)²
D2 = 100 × 0.64  =  64 cGy
```

Notice the dose did not drop by 20% just because the distance grew by 20% — it dropped by 36%, because of the squaring. This is also your friend in radiation protection: **stepping back a little buys a lot of safety** (more on that in Chapter 4).

Attenuation and the half-value layer

Distance is one way a beam weakens; passing through material is the other. As photons travel through tissue or shielding, some are absorbed or scattered away. This follows an **exponential** pattern: each equal thickness removes the same fraction of what is left, not the same amount.

The handiest measure of this is the **half-value layer (HVL)** — the thickness of a material that cuts the beam intensity in half.

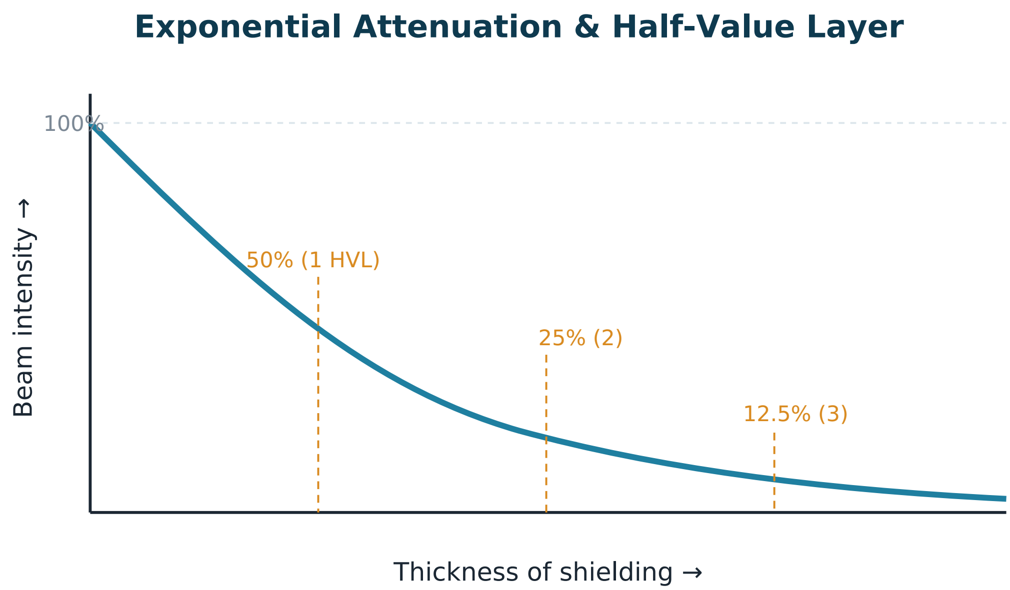

Figure 3.4 — Each half-value layer removes half of the remaining beam: 100% → 50% → 25% → 12.5%. Because it is always "half of what's left," the beam approaches zero but never quite gets there.

Worked example. A beam starts at 100%. If the HVL of a shielding material is 1.2 cm, how much gets through 3.6 cm?

```
3.6 cm ÷ 1.2 cm = 3 half-value layers
After 1 HVL: 50%
After 2 HVL: 25%
After 3 HVL: 12.5%
```

So 12.5% of the beam comes through. HVL is also a measure of **beam quality** (penetrating power): a higher-energy, more penetrating beam has a larger HVL because it takes more material to halve it.

> **Common mix-up:** Attenuation is **exponential**, not linear. Three HVLs do not block "150%" — each layer halves what remains, so you get 50% → 25% → 12.5%.

Part 3 · How photons interact with matter

When a photon does interact, it does so in one of three main ways. Knowing them explains image contrast, beam penetration, and why MV beams are skin-sparing.

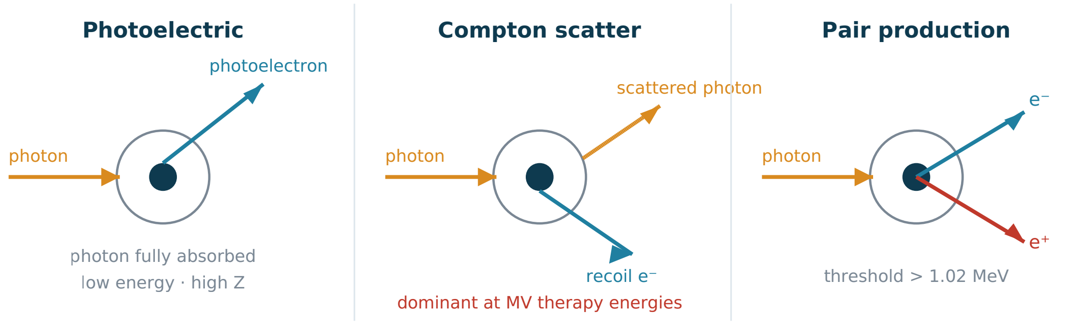

Figure 3.5 — The three photon interactions. Photoelectric: the photon is fully absorbed and an inner electron is ejected. Compton: the photon scatters off an outer electron, giving up part of its energy. Pair production: a high-energy photon converts into an electron–positron pair near the nucleus.

- **Photoelectric effect.** The photon hands **all** of its energy to a tightly bound inner-shell electron, which is ejected; no scattered photon survives. It dominates at **low energies** and is very sensitive to atomic number — roughly proportional to $\mathbf{Z^3}$. That strong Z dependence is exactly why **bone (high Z, calcium) shows up bright on kV images**: bone soaks up low-energy photons far more than soft tissue does. This is the physics behind the crisp skeletal detail you use when you bone-match a setup image.
- **Compton scatter.** The photon strikes a loosely bound **outer** electron, gives up part of its energy, and **scatters** off in a new direction while the electron recoils. Compton is almost independent of atomic number and **dominates across the megavoltage therapy range**. Two consequences follow: MV treatment beams deposit dose similarly in bone and soft tissue (so MV images have poor bone contrast), and scattered photons are the reason treatment rooms need **secondary** shielding (Chapter 4).
- **Pair production.** Only above a **threshold of 1.02 MeV**, a photon passing near a nucleus can convert into an **electron–positron pair** (matter created from energy). It becomes significant only at high energies and high Z, so it plays a minor role at common therapy energies.

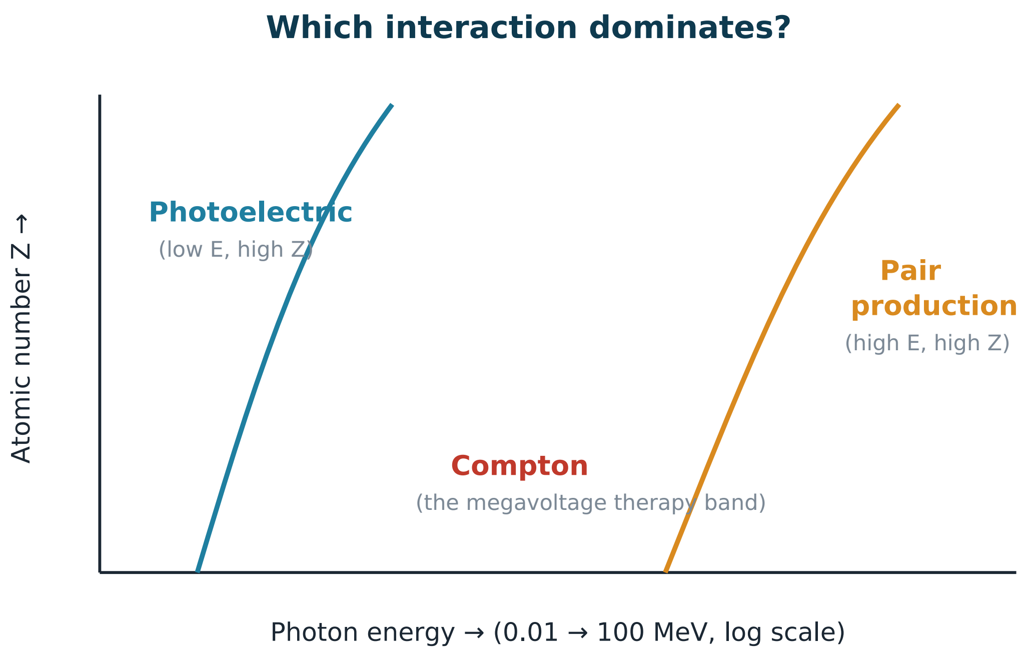

Figure 3.6 — Photoelectric rules at low energy/high Z, pair production at very high energy/high Z, and Compton owns the broad middle band — which is exactly where megavoltage therapy lives.

> **Key Point:** In the **megavoltage** beams you treat with every day, **Compton scatter dominates**. That single fact explains poor bone contrast on MV images, the need for scatter shielding, and the uniform way MV dose is deposited.

Build-up and skin-sparing

Here is a payoff of all that interaction physics. When an MV photon interacts, it sets fast electrons moving forward. Those electrons travel a short distance before they deposit their dose, so the maximum dose lands a little **below** the skin, not on it. The result is the **build-up region** and the skin-sparing that makes high-energy x-rays so useful.

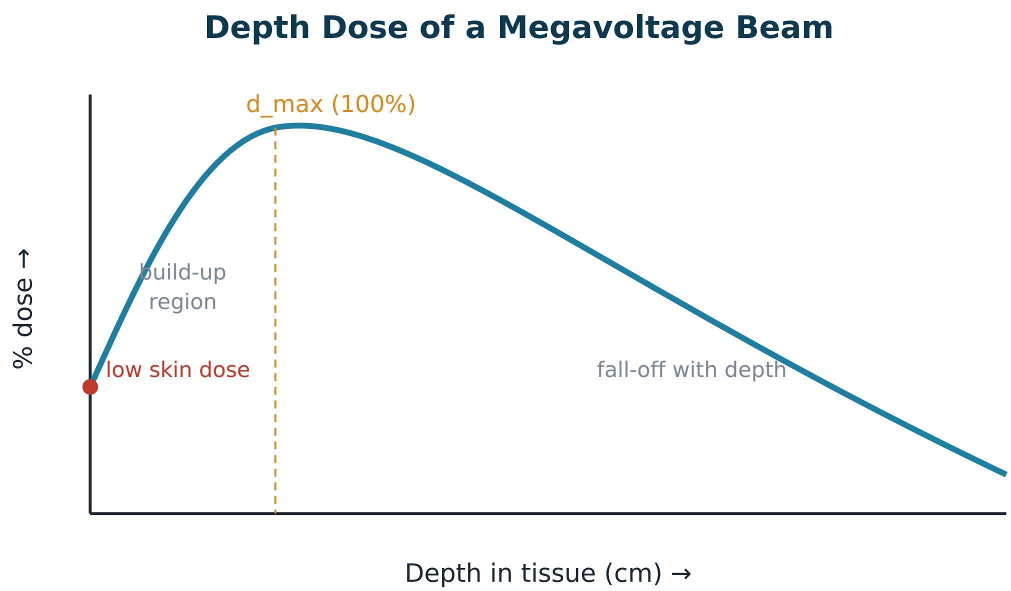

Figure 3.7 — A megavoltage beam enters with a low surface (skin) dose, builds up to its maximum at d_max, then falls off with depth. Higher energy pushes d_max deeper and spares the skin more.

The depth of maximum dose, **d_max**, gets deeper as energy rises (for example, roughly 1.5 cm for a 6 MV beam and around 3 cm for an 18 MV beam). This is why higher energies are chosen for deeper tumors — they put their peak dose where the target is, while keeping the skin dose comfortable.

Part 4 · Measuring radiation: the units

Three units cover almost everything on the exam.

Quantity	SI unit	Plain-English meaning	Older unit
Absorbed dose	**Gray (Gy)**	Energy actually deposited: 1 Gy = 1 joule per kilogram	rad (1 Gy = 100 rad)
Equivalent dose	**Sievert (Sv)**	Absorbed dose adjusted for how harmful the radiation type is	rem (1 Sv = 100 rem)
Activity	**Becquerel (Bq)**	How many atoms decay per second	curie (1 Ci = 3.7×10^{10} Bq)

A couple of anchors worth memorizing: **1 Gy = 100 rad** and **1 Sv = 100 rem**. For the x-rays, gammas, and electrons used in radiation therapy, the "harm factor" (radiation weighting factor) is 1, so **1 Gy of our beam equals 1 Sv**. The gray and sievert only diverge for high-LET radiations like alpha particles and neutrons.

> **Common mix-up:** **Gray** measures energy deposited in tissue — it is what the prescription is written in. **Sievert** measures biological risk and is the unit of dose limits and badge readings (Chapter 4). Same size for our beams, different jobs.

A quick word on calibration: physicists make sure "what the machine says" equals "what the patient gets" using the **AAPM TG-51** protocol, which calibrates photon beams at a reference depth of **10 cm in water** and specifies an electron beam's quality by **R_{50}** (the depth where dose falls to 50%). You do not perform this, but you should recognize the names.

Part 5 · Radiobiology: what radiation does to cells

Direct and indirect action

Radiation kills cells mainly by damaging **DNA**. It can do this two ways: **directly**, by breaking the DNA strand itself, or **indirectly**, by ionizing nearby water to create reactive **free radicals** that then attack the DNA. For the low-LET radiation we use most, the **indirect** path through water and free radicals causes the majority of the damage — which is also why **oxygen matters** so much (coming up).

The cell survival curve and the linear-quadratic model

If you irradiate a population of cells with rising doses and plot the fraction that survive on a log scale, you get the famous **cell survival curve**.

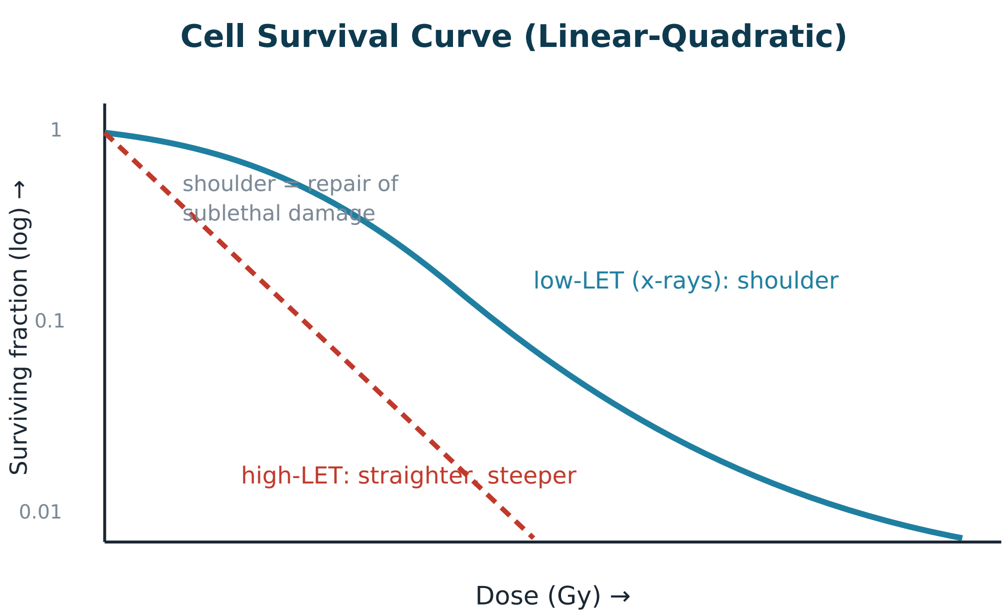

Figure 3.8 — For low-LET x-rays the curve has a "shoulder" at low dose (cells repairing sublethal damage) before bending steeply downward. High-LET radiation gives a straighter, steeper line — more lethal per gray, with little repair.

We describe the curve with the **linear-quadratic (LQ) model**:

> **Key Point: $S = e^{(-\alpha D - \beta D^2)}$** — survival falls off with a single-hit term (**α**, linear in dose) and a two-hit term (**β**, dose-squared).

The ratio **α/β** captures how a tissue responds to the size of each dose:

- **High α/β (≈ 10 Gy)** — tumors and early-responding tissues (skin, mucosa, gut). These care more about total dose.
- **Low α/β (≈ 3 Gy)** — late-responding tissues (spinal cord, lung, kidney). These are very sensitive to large doses per fraction.

This difference is the whole reason we usually give many **small** fractions: small daily doses spare the low-α/β late-responding normal tissues while still controlling the tumor.

LET and RBE

- **LET (Linear Energy Transfer)** is how densely a radiation deposits energy along its track. X-rays and electrons are **low-LET** (sparse ionizations); alpha particles, neutrons, and carbon ions are **high-LET** (dense ionizations that overwhelm repair).
- **RBE (Relative Biological Effectiveness)** compares how much of a reference radiation it takes to get the same effect as a test radiation. High-LET radiation has **RBE > 1** — it does more biological damage per gray.

> **Common mix-up:** LET describes the radiation's track structure; RBE describes the biological result. As LET rises, RBE generally rises with it — up to a point.

The oxygen effect

Because so much damage is done indirectly through free radicals, **oxygen** dramatically increases radiation's killing power: it "fixes" (makes permanent) the chemical damage. Well-oxygenated cells are far more radiosensitive than hypoxic (oxygen-starved) ones.

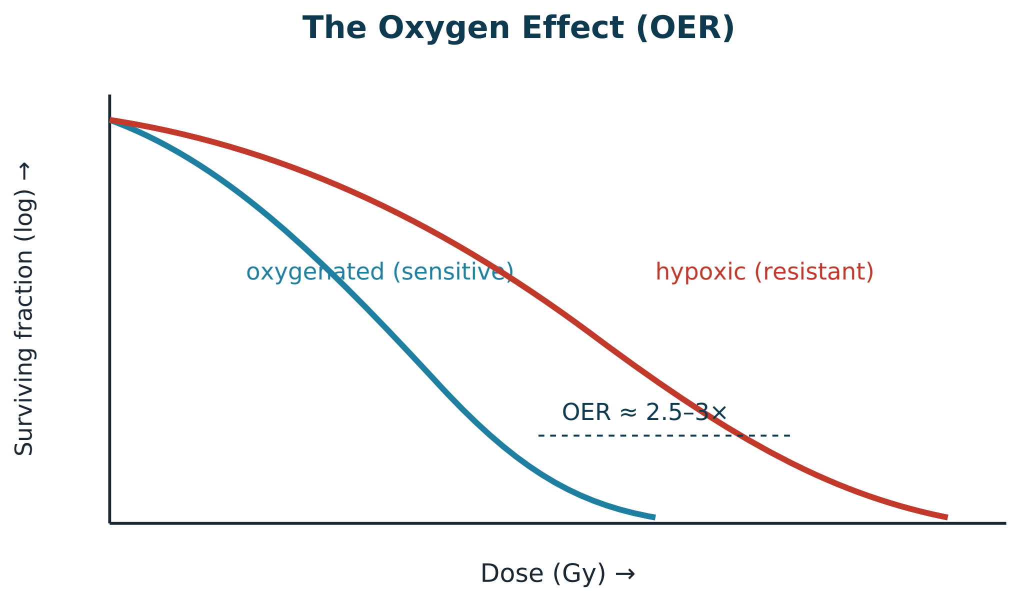

Figure 3.9 — Oxygenated cells (left curve) die at lower doses than hypoxic cells (right curve). The ratio of doses needed for the same effect is the Oxygen Enhancement Ratio, about 2.5–3 for low-LET radiation.

We quantify this with the **Oxygen Enhancement Ratio (OER) ≈ 2.5–3** for x-rays. Tumors often contain hypoxic, resistant pockets — and this is one more reason fractionation helps: between fractions, tumors **reoxygenate**, turning resistant cells into sensitive ones.

The Four R's of radiobiology

These four processes explain why fractionation works:

1. **Repair** — between fractions, normal cells repair sublethal damage (the "shoulder" returns each day). Late-responding normal tissue benefits most.
2. **Repopulation** — surviving cells divide during a long course; this is why dragging treatment out too long can let a tumor regrow.
3. **Reoxygenation** — hypoxic tumor regions regain oxygen between fractions and become easier to kill.
4. **Reassortment (redistribution)** — cells get caught in radiosensitive phases of their cycle over repeated fractions.

> **Key Point:** Repair and Repopulation tend to **protect** tissues; Reoxygenation and Reassortment tend to **sensitize** the tumor. Fractionation is the art of tilting that balance in the patient's favor.

Fractionation math: BED and EQD2

To compare different fraction schemes fairly, we use **Biologically Effective Dose (BED)** and the **Equivalent Dose in 2-Gy fractions (EQD2)**:

```
BED  = n · d · (1 + d / (α/β))
EQD2 = BED / (1 + 2 / (α/β))
   n = number of fractions
   d = dose per fraction
```

Worked example 1 — a standard course, two tissues. 60 Gy in 30 fractions (so n = 30, d = 2 Gy).

```
Tumor (α/β = 10):       BED = 60 × (1 + 2/10) = 60 × 1.2   = 72 Gy
Late tissue (α/β = 3):  BED = 60 × (1 + 2/3)  = 60 × 1.667 ≈ 100 Gy
```

The same physical 60 Gy is biologically "heavier" for the late-responding tissue — which is precisely why we respect its dose limits.

Worked example 2 — converting an SBRT course to EQD2. A lung tumor gets 50 Gy in 5 fractions (n = 5, d = 10 Gy, α/β = 10).

```
BED  = 5 × 10 × (1 + 10/10) = 50 × 2 = 100 Gy
EQD2 = 100 / (1 + 2/10)     = 100 / 1.2 ≈ 83 Gy
```

So that short, punchy SBRT course is biologically like ~83 Gy given in gentle 2-Gy fractions — far more than its 50 Gy "sticker price." This is why high-dose-per-fraction treatments are so potent (and why their normal-tissue limits are handled carefully).

Part 6 · Effects on tissue and people

Deterministic versus stochastic effects

Radiation's harmful effects come in two flavors, and the exam loves to test the distinction.

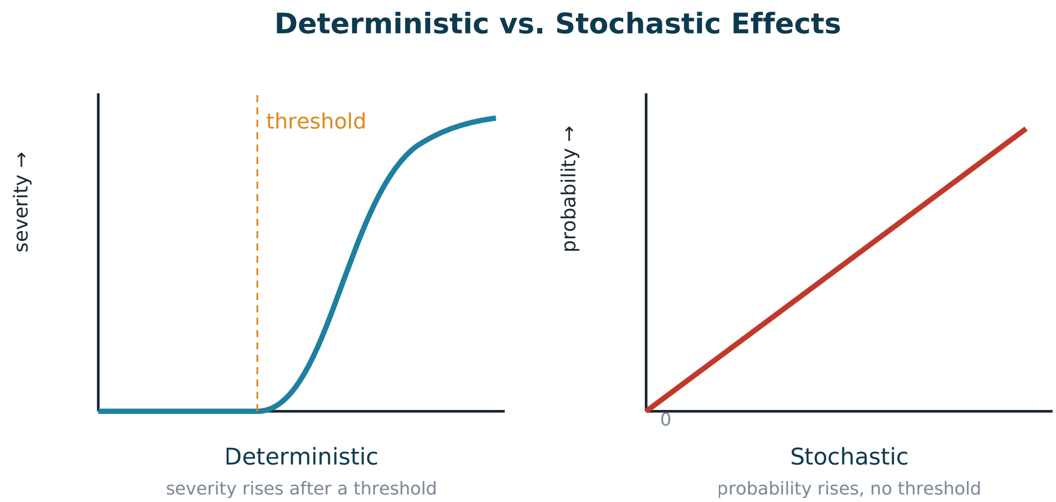

Figure 3.10 — Deterministic effects have a threshold: below it, nothing; above it, severity grows with dose. Stochastic effects have no threshold: the probability rises with dose, but severity does not.

- **Deterministic (tissue reactions)** — there is a **threshold dose**, and above it the **severity** climbs with dose. Examples: skin erythema, cataracts, sterility, fibrosis. These are predictable and dose-dependent.
- **Stochastic (chance) effects — no threshold**; the **probability** of the effect rises with dose, but the severity does not depend on dose. Examples: radiation-induced cancer and heritable effects. This is the basis of the "no safe dose" precaution behind ALARA.

Common mix-up: Deterministic = severity rises with dose, above a threshold. Stochastic = probability rises with dose, no threshold. A cataract is deterministic; a radiation-induced cancer is stochastic.

Normal-tissue tolerance

Every organ has a dose it can tolerate before complications become likely. A common shorthand is **$TD_{5/5}$** — the dose expected to cause a 5% complication rate within 5 years. A few values worth recognizing (conventional fractionation):

- **Spinal cord:** about **45–50 Gy** (we keep cord dose well under this to avoid myelopathy).
- **Lung:** mean dose roughly **≤ 20 Gy**, or keep the volume getting ≥ 20 Gy (V20) under about 35%, to limit pneumonitis.
- **Eye lens:** a threshold around **0.5 Gy** for cataract formation.

These numbers are why the prescription and plan obey **organ-at-risk constraints**, and why the dose-volume histogram (Chapter 6) is checked so carefully.

Check yourself

1. Cobalt-60 emits gamma rays of what energies, and what is its half-life? 1.17 and 1.33 MeV (average ≈ 1.25 MeV), with a half-life of 5.27 years.

2. Which photon interaction dominates at megavoltage therapy energies, and name one practical consequence. Compton scatter. Consequences include poor bone contrast on MV images and the need for secondary (scatter) shielding.

3. A source gives 200 cGy at 100 cm. Using the inverse-square law, what is the dose rate at 140 cm? $D_2 = 200 \times (100/140)^2 = 200 \times 0.510 \approx 102$ cGy.

4. A beam is 100%. After passing through 2 half-value layers, how much remains? 25% (100% → 50% → 25%). Attenuation is exponential — each HVL halves what is left.

5. What is the difference between the gray and the sievert? The gray measures absorbed dose (energy deposited, used for prescriptions); the sievert measures equivalent dose (biological risk, used for dose limits). For therapy x-rays they are numerically equal.

6. Convert 50 Gy in 5 fractions (α/β = 10) to BED. $BED = 5 \times 10 \times (1 + 10/10) = 100$ Gy.

7. A cataract from radiation is which type of effect — deterministic or stochastic — and why? Deterministic: it has a threshold dose, and severity increases with dose above that threshold.

8. Why does fractionation help spare late-responding normal tissue? Late-responding tissue has a low α/β (~3 Gy), making it very sensitive to large doses per fraction; small daily fractions plus overnight Repair keep its biological dose low while the tumor is still controlled.

Chapter references

Free, full-text sources

1. IAEA — Radiation Oncology Physics: A Handbook for Teachers and Students (free PDF; beam production, interactions, attenuation, depth dose, units, radiobiology): https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1196_web.pdf.
2. AAPM Reports — TG-51 reference dosimetry (R_{50} for electron beam quality), free: https://www.aapm.org/pubs/reports/.
3. U.S. NRC — 10 CFR Part 20, Standards for Protection Against Radiation (quantities, units): https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/.
4. CDC — Radiation and your health (deterministic vs stochastic effects, ALARA): https://www.cdc.gov/radiation-health/.

For deeper reading (library / textbook)

1. Khan FM, Gibbons JP. The Physics of Radiation Therapy. Wolters Kluwer.
2. Hall EJ, Giaccia AJ. Radiobiology for the Radiologist. Wolters Kluwer.

This chapter offers original educational explanations. It is not affiliated with or endorsed by the ARRT and does not reproduce actual exam questions.

Chapter 4 — Radiation Protection, Equipment Operation & Quality Assurance

What you'll learn

- How the **ALARA** principle works and how its three tools — **time, distance, and shielding** — actually lower dose, including why distance is so powerful (the inverse-square law).
- The U.S. **NRC dose limits** for occupational workers, the public, the declared-pregnant worker, and specific tissues — straight from 10 CFR 20.
- How treatment rooms are shielded: **primary vs. secondary barriers** and the design factors **W**, **U**, and **T**.
- How **personnel dosimeters** (TLD, film badge, OSL) work, where you wear them, and how often they're read.
- The major **linear accelerator components** and the **safety interlocks** that protect patients and staff.
- The backbone of a **quality assurance (QA)** program — linac checks (TG-142), imaging/IGRT checks (TG-179), and CT-simulator checks (TG-66).

Why this matters

Radiation therapy delivers enormous, deliberate doses to patients while keeping the people around the beam safe. That balance only works because of layered protection: smart habits, well-designed rooms, monitored dosimeters, and machines that are checked constantly. As a therapist, you are the daily guardian of all four — and the ARRT registry will expect you to know the numbers and the reasons behind them cold.

The ALARA Principle

ALARA stands for **As Low As Reasonably Achievable**. It is the guiding philosophy of all radiation protection. The idea is simple: even when a dose is legally allowed, you still try to keep it as low as you reasonably can, factoring in technology, cost, and the benefit of the procedure.

Think of ALARA like driving under the speed limit. The limit is the maximum, not a target. A careful driver routinely goes slower than allowed when conditions call for it. ALARA asks you to treat every dose limit the same way — as a ceiling you stay well beneath.

ALARA rests on three practical tools you can adjust at the bedside or in the control area: **time, distance, and shielding**.

> **Key Point:** ALARA is not a dose limit — it is a mindset. Limits are legal ceilings; ALARA pushes you to stay far below them whenever practical.

Tool 1 — Time

The less time you spend near a radiation source, the less dose you receive. Dose accumulates in direct proportion to exposure time: double your time in the field, and you roughly double your dose.

In practice this means working efficiently and confidently around active sources — for example, during **brachytherapy** (treatment using radioactive sources placed in or near the tumor) — and never lingering. Rehearsing a procedure so you move quickly and deliberately is itself a radiation-protection strategy.

Tool 2 — Distance and the Inverse-Square Law

Distance is the single most powerful protection tool, and the reason is the **inverse-square law**. This law states that the intensity of radiation from a point source falls off with the square of the distance from that source.

In formula form:

$$\mathbf{I_1 / I_2 = (d_2)^2 / (d_1)^2}$$

where I is intensity and d is distance. The practical takeaway: if you **double** your distance from a source, the intensity drops to **one-quarter** (because $2^2 = 4$). Triple the distance, and intensity drops to one-ninth ($3^2 = 9$).

Picture radiation spreading out from a source like paint sprayed from a nozzle. Up close, the paint is dense on a small patch. As the spray fans out, the same amount of paint covers a much larger area, so any one spot gets far less. Stepping back a single extra step buys you outsized protection.

> **Common mix-up:** Students often think doubling distance halves dose. It does not — it cuts dose to a **quarter**. The relationship is squared, not linear.

Tool 3 — Shielding

Shielding places absorbing material between you and the source. **High-density, high-atomic-number materials** like **lead** stop photons most efficiently per unit thickness, while **concrete** is used in bulk for walls because it is cheap and structural.

The amount of shielding is often described in **half-value layers (HVL)** — the thickness of material needed to cut the beam intensity in half. Stack enough half-value layers and the beam is reduced by a large, predictable factor.

In therapy, shielding shows up as the lead in the linac head, the thick concrete-and-lead vault walls, and the lead-lined door — not as a lead apron. (Lead aprons help in low-energy diagnostic imaging but are useless against the high-energy beams of a linac, which would pass right through them.)

Occupational and Public Dose Limits (U.S. NRC, 10 CFR 20)

The U.S. Nuclear Regulatory Commission (**NRC**) sets enforceable dose limits in federal regulation **10 CFR Part 20** [1]. These are the numbers used in the United States and the ones the registry expects.

A quick unit note: the **sievert (Sv)** and **millisievert (mSv)** are SI units of effective dose (dose adjusted for biological harm). The older U.S. unit is the **rem**, where **1 rem = 10 mSv** (so 1 Sv = 100 rem).

Category	Limit (SI)	Limit (conventional)
Occupational whole-body (annual)	**50 mSv/year**	5 rem/year
Occupational **cumulative** lifetime	**10 mSv × age in years**	1 rem × age
Lens of the eye (annual)	**150 mSv/year**	15 rem/year
Skin / extremities (annual)	**500 mSv/year**	50 rem/year
Public (annual)	**1 mSv/year**	100 mrem/year
Declared-pregnant worker — embryo/fetus (entire pregnancy)	**5 mSv total** (~0.5 mSv/month)	0.5 rem

A few clarifications:

- The **whole-body** occupational limit is **50 mSv (5 rem) per year**. The NRC also applies a **cumulative** limit of **10 mSv multiplied by your age in years** — so a 40-year-old worker's lifetime total should not exceed 400 mSv.
- The **public** limit is **1 mSv (100 mrem) per year** — fifty times stricter than the occupational limit, because the public hasn't accepted the risk as part of a job and isn't monitored.
- A worker who voluntarily informs her employer in writing that she is pregnant becomes a **declared-pregnant worker**. The dose limit to the **embryo/fetus** is **5 mSv (0.5**

rem) over the entire gestation, and the NRC asks employers to avoid large monthly spikes — roughly **0.5 mSv per month**.

- The **lens of the eye** limit is **150 mSv/year**, and **skin and extremities** are limited to **500 mSv/year**. These tissues tolerate more local dose than the whole body.

> **Key Point:** Memorize the headline set: **occupational 50 mSv/yr, public 1 mSv/yr, fetus 5 mSv total, cumulative 10 mSv × age.** These four are registry favorites.
> **Common mix-up:** Some review sources quote the international **ICRP** occupational figure of 20 mSv/year or a 0.1 mSv public figure. For the U.S. registry, use the **NRC 10 CFR 20** values above — **50 mSv** occupational and **1 mSv** public.

Facility Shielding (NCRP Report 151)

Treatment rooms are housed in heavily shielded vaults. The design rules come from **NCRP Report 151** [2]. Shielding barriers fall into two types based on what radiation they intercept.

Primary vs. Secondary Barriers

- A **primary barrier** is the wall (or floor/ceiling) that the **direct, useful beam** can point at. Because it faces the full-strength beam, it is the thickest part of the vault.
- A **secondary barrier** handles only **leakage** (radiation that escapes the linac head despite its shielding) and **scatter** (radiation that bounces off the patient and room surfaces). These are weaker, so secondary barriers can be thinner.

Picture the primary barrier as the backstop directly behind home plate at a batting cage — it takes the hardest hits. Secondary barriers are the side netting that only catches deflections.

The Design Factors: W, U, and T

NCRP-151 calculates required barrier thickness using three factors:

Factor	Name	What it means
W	Workload	How much the machine is used — total dose delivered at 1 meter per week (e.g., Gy/week). Busier rooms need more shielding.
U	Use factor	The fraction of the workload the beam is aimed at a particular barrier. A floor might have U near 1; a given wall might be ¼.
T	Occupancy factor	How occupied the space on the other side of the barrier is. A full-time office (T = 1) demands more shielding than a rarely used storage closet or parking lot (T as low as 1/40).

These combine so that a barrier protecting a busy office, frequently in the beam's path, in a high-workload room, must be thickest. Vault walls are typically built of **concrete** in bulk, with **lead** added where space is tight (lead gives more attenuation per inch).

Controlled vs. Uncontrolled Areas

- A **controlled area** is occupied mainly by trained, monitored radiation workers (the control console area). It is held to the occupational framework.
- An **uncontrolled area** is anywhere the general public may be — waiting rooms, hallways, the floor above. These must meet the **public** limit of **1 mSv/year**, which is why shielding toward them is so demanding.

Personnel Dosimetry

A **dosimeter** is a small device a worker wears to measure accumulated occupational dose. It does not protect you — it records what you received, much like a car's odometer logs distance after the fact.

Type	How it works	Key traits
TLD (thermoluminescent dosimeter)	Crystals (commonly **lithium fluoride**) trap energy from radiation; heating them later releases light proportional to dose	**Reusable**; accurate; no instant readout
Film badge	Radiation darkens photographic film inside the badge	Provides a **permanent physical record**; **single-use** (not reusable); sensitive to heat/humidity
OSL (optically stimulated luminescence)	Aluminum-oxide crystals trap energy; **laser light** (not heat) releases it for reading	Reusable; can be re-read; very sensitive

Dosimeters are typically **read monthly** by an outside service that reports each worker's dose.

Placement matters:

- Worn at the **collar** (outside any apron) for staff in diagnostic settings, estimating dose to the head, neck, and lens.
- Worn at the **waist** for whole-body monitoring.
- A pregnant worker often wears a **second** dosimeter at the **waist/abdomen** to track fetal dose specifically.

> **Key Point:** TLD and OSL are **reusable**; the **film badge is single-use** but leaves a **permanent record**. That trade-off — reusability vs. a permanent archive — is the classic exam distinction.

Linear Accelerator Components

A medical **linear accelerator (linac)** speeds up electrons to high energy and either uses them directly or converts them into a photon (x-ray) beam. Knowing the path the beam takes through the head helps you understand both treatment and QA.

Following the beam from start to patient:

1. **Electron gun** — injects electrons into the accelerating structure, where microwaves boost them to high energy.
2. **Tungsten target** — for **photon** treatments, the fast electrons slam into this dense metal target, producing x-rays (bremsstrahlung). For **electron** treatments, the target is moved out of the way.
3. **Primary collimator** — a fixed shielding aperture that defines the largest possible beam and blocks stray radiation.
4. **Flattening filter** — a cone-shaped piece of metal that evens out the beam, which is naturally peaked in the center, into a uniform "flat" profile across the field. - **Flattening-filter-free (FFF)** modes remove this filter for very high dose rates, used in SRS/SBRT where the beam is small and speed reduces patient motion. The profile is then peaked but corrected by the planning system.
5. **Dual monitor ionization chambers** — two independent chambers measure the dose being delivered in real time. The redundancy is a safety feature: if one fails or they disagree, the beam shuts off.
6. **Jaws** — movable lead/tungsten blocks that set the rectangular field size.
7. **Multileaf collimator (MLC)** — dozens of thin, independently driven leaves that shape the beam to the tumor's outline and enable intensity-modulated and dynamic treatments.

> **Common mix-up:** The **target** makes photons; remove it and you have an electron beam. The **flattening filter** flattens photon beams; removing it gives the high-dose-rate **FFF** mode. Two different parts, two different jobs.

Safety Interlocks

An **interlock** is an automatic safety circuit that stops the beam (or prevents it starting) when a condition is unsafe. Interlocks are deliberately **redundant** — built with backup circuits — so that no single failure can defeat them.

Key interlocks include:

- **Door interlock** — the beam cannot run if the vault door is open, keeping people out of the active field.
- **Beam-off / emergency-off (E-stop)** — large red buttons in the room and at the console that immediately kill the beam and machine motion.
- **Motion-disable / collision interlocks** — stop gantry, couch, or collimator movement if a collision or out-of-range condition is detected.

> **Key Point:** If an interlock trips, the safe response is to **stop, identify the cause, and resolve it** — never to bypass or "force" the machine past a safety circuit.

Linac Quality Assurance (AAPM TG-142)

Quality assurance (QA) is the routine testing that confirms the machine performs within tolerance. The standard reference for linac QA is **AAPM Task Group 142 (TG-142)** [3]. Tests are grouped by how often they're done.

Daily QA (done every treatment day)

- **Output constancy** — the dose delivered matches the expected value, typically within about **±3%**.
- **Beam symmetry and flatness** — the beam is even and centered.
- **Lasers** — the room positioning lasers point at the correct isocenter.
- **Door and audiovisual interlocks** — confirmed working before patients are treated.

Monthly QA

More detailed checks of output, energy, field-size accuracy, MLC positioning, and mechanical alignment.

Annual QA

The most thorough review — absolute dose calibration, full mechanical and dosimetric characterization, and imaging-system verification.

> **Key Point: Tolerances tighten with technique.** Conventional treatments allow looser margins; **IMRT** (intensity-modulated radiation therapy) and especially **SRS/SBRT** (stereotactic radiosurgery / body radiotherapy) demand much tighter ones, because tiny errors matter more when small, high-dose fields sit next to critical structures.

Imaging-System QA (AAPM TG-179)

Modern treatment relies on **image-guided radiation therapy (IGRT)** — using on-board imaging like **kV** (kilovoltage) planar images or **CBCT** (cone-beam CT) to verify the patient's position right before treatment. **AAPM TG-179** covers QA for these systems [4].

The single most important check: the **imaging isocenter must agree with the treatment isocenter within about 1 mm**. If the picture you align to is even slightly offset from where the beam actually goes, you would shift the patient to the wrong place — a precise image in the wrong spot is worse than no image.

Image-quality metrics are also tracked, including:

- **HU / CT-number linearity** — Hounsfield Units (the CT density scale) read correctly across materials.
- **Spatial resolution / MTF** — the **modulation transfer function** describes how well fine detail is preserved; it tells you the smallest features the system can resolve.

> **Key Point:** Imaging-to-treatment isocenter agreement within **~1 mm (TG-179)** is the headline imaging-QA number. Geometry first; image quality second.

CT-Simulator QA (AAPM TG-66)

The **CT simulator** is the CT scanner used to plan treatment — it captures the patient's anatomy in the treatment position and defines the coordinate system the whole plan is built on. **AAPM TG-66** specifies its QA [5].

Critical CT-sim checks:

- **Laser and geometry accuracy** — the external positioning lasers and the scan geometry must be precise, because they set the patient's reference marks (tattoos) and the planning origin.
- **CT-number (HU) calibration** — the scanner must read **water = 0 HU** and **air = −1000 HU**. These two anchor points keep the density scale honest, which matters because the planning system converts HU into the electron densities used for dose calculation.

If the CT simulator's geometry or HU calibration drifts, **every plan built from it inherits the error** — which is why TG-66 treats the simulator as a foundation that must be verified, not assumed.

Chapter summary

Radiation protection is a system of layers. **ALARA** sets the mindset; **time, distance, and shielding** are the levers, with distance amplified by the **inverse-square law**. The **NRC's 10 CFR 20** limits — 50 mSv/yr occupational, 1 mSv/yr public, 5 mSv to the embryo/fetus, 10 mSv × age cumulative — define the legal ceilings, and **NCRP-151** shielding (primary vs. secondary barriers, sized by **W, U, T**) keeps everyone beneath them. **Dosimeters** record what slips through. Finally, the machines themselves are kept honest by **interlocks** and a tiered QA program — **TG-142** for the linac, **TG-179** for imaging, **TG-66** for the CT simulator. Master these and you can answer most protection-and-equipment questions on sight.

Check yourself

1. You are standing 1 meter from a small radioactive source. If you step back to 2 meters, how does your dose rate change? It drops to one-quarter. By the inverse-square law, doubling the distance (2^2) reduces intensity by a factor of four — not by half.

2. What are the NRC annual dose limits for an occupational whole-body worker and for a member of the public? Occupational whole-body is 50 mSv/year (5 rem). The public limit is 1 mSv/year (100 mrem) — fifty times stricter.

3. What is the dose limit to the embryo/fetus of a declared-pregnant worker, and over what period? 5 mSv (0.5 rem) over the entire gestation, with an effort to keep it roughly uniform at about 0.5 mSv per month.

4. A vault wall faces the direct treatment beam. Is it a primary or secondary barrier, and which NCRP-151 factor accounts for how often the beam points at it? It is a primary barrier (it intercepts the useful beam). The use factor U accounts for the fraction of the workload aimed at that specific barrier.

5. Which dosimeters are reusable, and which leaves a permanent physical record? TLD and OSL are reusable. The film badge is single-use but provides a permanent physical record.

6. Per TG-179, how closely must the imaging isocenter match the treatment isocenter, and why does it matter? Within about 1 mm. If the image you align the patient to is offset from where the beam actually delivers, you would position the patient incorrectly despite a "good" image.

Chapter references

Free, full-text sources

1. U.S. NRC — 10 CFR Part 20, Standards for Protection Against Radiation (occupational, public, and embryo/fetus dose limits): https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/.
2. AAPM Reports — linac QA (TG-142), IGRT/CBCT QA (TG-179), CT-simulator QA (TG-66), all free: https://www.aapm.org/pubs/reports/.
3. CDC — ALARA and radiation safety: https://www.cdc.gov/radiation-health/.
4. IAEA — *Radiation Oncology Physics: A Handbook for Teachers and Students* (free PDF; shielding, equipment, QA chapters): https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1196_web.pdf.

For deeper reading (library / standards)

1. NCRP Report 151 — Structural shielding design for megavoltage radiotherapy facilities (NCRP; purchase).
2. Khan FM, Gibbons JP. *The Physics of Radiation Therapy*. Wolters Kluwer.

Chapter 5 — Treatment Sites & Tumors

What you'll learn

- A shared framework for thinking about any cancer: what it is made of (histology), how aggressive it looks (grade), how far it has spread (stage), and how it travels (routes of spread).
- Why radiation dose and fractionation change from site to site, and why no single number is ever "the" correct dose.
- The key anatomy, common tumor types, staging logic, and nodal drainage patterns for the major treatment regions you will see in the clinic.
- Commonly used dose/fractionation patterns for each site, framed as typical examples rather than fixed rules.
- The organs at risk (OARs) that quietly shape every plan, and special techniques (like breath-hold or fiducial markers) that exist because of them.

Why this matters

As a radiation therapist, you treat a patient, not a diagnosis on a chart. But the diagnosis tells you almost everything about why the plan looks the way it does: where the beams point, how big the margins are, how many days of treatment, and which structures the dosimetrist fought hardest to spare. When you understand the disease, the plan stops looking like arbitrary numbers and starts looking like a logical argument. This chapter builds that understanding region by region.

The shared framework (read this first)

Before we tour the body, let's build a mental toolkit. Almost every cancer is described using the same four ideas.

Histology means "what the tumor is made of," seen under a microscope. The suffix usually tells you the tissue of origin. A carcinoma arises from epithelial (lining/surface) tissue and is by far the most common kind. A sarcoma arises from connective tissue (bone, muscle, fat). A lymphoma arises from lymphatic tissue, and a blastoma (often pediatric) arises from immature precursor cells. Within carcinomas, squamous cell carcinoma (SCC) comes from flat lining cells and adenocarcinoma comes from glandular cells.

Grade describes how abnormal the cells look. Well-differentiated (low-grade) cells still resemble the normal tissue and tend to behave less aggressively. Poorly differentiated (high-grade) cells look chaotic and tend to grow and spread faster. Think of grade as the tumor's "personality" and stage as its "location report."

Stage describes how far the cancer has spread. The dominant system is the **AJCC TNM** system (American Joint Committee on Cancer), now in its 8th edition [1]:

Letter	Question it answers	General idea
T	How big/invasive is the primary **T**umor?	Higher number = larger or deeper
N	Are regional lymph **N**odes involved?	N0 = none; N1–N3 = increasing nodal burden
M	Is there distant **M**etastasis?	M0 = none; M1 = spread to distant organs

These T, N, and M values are combined into an overall **stage group** (I–IV). The TNM details differ for every site, so the same "T2" means different things in the larynx than in the prostate.

Routes of spread describe how a cancer travels, which explains where therapists aim:

- **Local (direct) extension** — the tumor grows outward into neighboring tissue. This is why we draw a margin around the visible disease.
- **Lymphatic spread** — cells travel through lymph channels to regional nodes. This is why we often treat nodal regions that look clinically normal but are at risk. Each site drains to predictable node groups.
- **Hematogenous spread** — cells enter the bloodstream and seed distant organs. Sarcomas famously do this to the lung; many carcinomas seed bone, liver, lung, and brain.

Finally, a rule that holds for the entire chapter: **doses vary by protocol.** Every regimen below is a commonly used example drawn from NCCN guidelines and standard clinical references [2][3][4]. Institutions differ, trials evolve, and the prescribing physician owns the final number. Learn the pattern (why SRS is one big fraction, why palliation is short), not the digits.

> **Key Point:** When a dose surprises you, ask "what is the intent?" Cure (definitive) regimens are higher and longer; palliative regimens are lower and shorter; ablative stereotactic regimens are very high dose in very few fractions to small, well-targeted volumes.

Central nervous system (brain)

Anatomy. The brain sits inside the rigid skull, so even small swelling causes symptoms. The blood–brain barrier limits many drugs, which raises the role of radiation. Critical OARs include the optic chiasm, optic nerves, brainstem, cochlea, and hippocampus.

Common histology. Glioblastoma (GBM) is the most common malignant primary brain tumor in adults — a high-grade (WHO grade 4) glioma that infiltrates far beyond what you can see on imaging. Vestibular schwannoma (also called acoustic neuroma) is a benign tumor of the nerve sheath on cranial nerve VIII. Brain metastases are far more common than primary brain tumors and arrive hematogenously from lung, breast, melanoma, and others.

Staging notes. Gliomas are not staged by TNM; they are classified by WHO grade and molecular markers. Metastases are by definition stage IV disease of their primary.

Routes of spread. GBM spreads by local infiltration along white-matter tracts; it almost never goes to nodes or outside the CNS. That infiltrative habit is exactly why the GBM target volume is generous.

Commonly used regimens.

- Glioblastoma: a commonly used regimen is **60 Gy in 30 fractions** with concurrent and adjuvant **temozolomide** (an oral chemotherapy) [2][5]. The clinical target volume (CTV) typically covers the contrast-enhancing tumor and resection cavity **plus the surrounding FLAIR (edema) region** with a margin, because microscopic disease hides in that edema.
- Vestibular schwannoma: often treated with **stereotactic radiosurgery (SRS)** — a single highly focused dose, e.g., **~12–13 Gy in one fraction** — to halt growth and preserve hearing [3].
- Brain metastases: **SRS** for a limited number of lesions, versus **whole-brain radiation therapy (WBRT)** for diffuse disease, e.g., **30 Gy in 10 fractions** [2].

> **Common mix-up:** A schwannoma is benign, yet we use SRS — a very high single dose. Remember that "radiosurgery" describes the technique (one or few precise fractions), not the malignancy. Benign and malignant targets can both be ablated.

Head and neck

Anatomy. This region packs the oral cavity, oropharynx, nasopharynx, larynx, salivary glands, and a dense network of lymph nodes into a small space. Cervical (neck) nodes are mapped into **levels I–VI**, a numbering you will see on nearly every head-and-neck plan.

Common histology. The vast majority are **squamous cell carcinoma** of the mucosal lining.

Staging notes. HPV status changed staging. HPV (human papillomavirus)-positive oropharyngeal SCC has a much better prognosis and gets its own staging table in AJCC

8th edition, separate from HPV-negative disease [1]. **Nasopharyngeal carcinoma** is strongly associated with **Epstein–Barr virus (EBV)**.

Routes of spread. Predominantly local and **lymphatic** to the cervical node levels, in patterns that depend on the primary site. This is why elective nodal irradiation is so common here.

Commonly used regimens.

- Definitive SCC (e.g., oropharynx): a commonly used regimen is **~70 Gy in 35 fractions** with concurrent **cisplatin** chemotherapy [2][3].
- Nasopharynx: treated with **IMRT** (intensity-modulated radiation therapy) to wrap dose around the skull base and spare salivary glands, with chemotherapy.
- Early glottic larynx (a small vocal-cord cancer): a commonly used regimen is **~63–66 Gy** with a focus on **voice preservation** — the larynx is treated, surgery is avoided, and speech is spared.

> **Key Point:** When you see elective treatment of normal-looking neck nodes, the disease almost certainly spreads lymphatically. The node levels treated are a fingerprint of where the primary drains.

Thorax (lung)

Anatomy. The lungs surround the heart, esophagus, and spinal cord; the central "mediastinum" holds great vessels and nodes. Lung tissue moves with breathing, which drives motion-management techniques.

Common histology. Two big families: **non-small-cell lung cancer (NSCLC)** — adenocarcinoma and SCC — accounts for ~85%, and **small-cell lung cancer (SCLC)** is a fast, aggressive neuroendocrine type.

Staging notes. NSCLC uses TNM. SCLC is often described simply as **limited-stage** (fits in one tolerable radiation field) or **extensive-stage**.

Routes of spread. Local invasion, lymphatic spread to mediastinal/hilar nodes, and hematogenous spread (brain is a notorious destination — which is why PCI exists, below).

Commonly used regimens.

- Definitive NSCLC (locally advanced): a commonly used regimen is **~60 Gy** with concurrent **chemotherapy** [2].
- Peripheral early-stage NSCLC: **SBRT** (stereotactic body radiation therapy), e.g., **50 Gy in 5 fractions** — high dose, few fractions, to a small mobile target. Motion management matters intensely here.

- Limited-stage SCLC: concurrent **chemoradiation**, followed in responders by **prophylactic cranial irradiation (PCI)**, e.g., **~25 Gy in 10 fractions**, to treat microscopic brain disease before it appears [2].

> **Common mix-up:** PCI treats a brain that looks normal on the scan. We irradiate it because SCLC seeds the brain so reliably that prevention beats waiting. It is prophylactic, not therapeutic.

Breast

Anatomy. The breast sits over the chest wall, ribs, and — on the left — close to the heart. Regional nodes follow three groups: **axillary levels I–III**, the **internal mammary** chain (along the sternum), and the **supraclavicular** nodes above the clavicle.

Common histology. Invasive ductal carcinoma (now often called invasive carcinoma of no special type) is most common; invasive lobular is next.

Staging notes. AJCC 8th edition added **biomarkers** (ER, PR, HER2) and grade to anatomic TNM, creating a "prognostic stage" [1]. **Inflammatory breast cancer** is **T4d** — a distinct, aggressive presentation with skin involvement.

Routes of spread. Local, then lymphatic (axilla first), then hematogenous to bone, lung, liver, brain.

Commonly used regimens.

- Whole-breast after lumpectomy: a commonly used regimen is **50 Gy in 25 fractions**, or **hypofractionated 40–42.5 Gy in 15 fractions**, frequently with a **boost** to the tumor bed [2][3].
- Inflammatory or node-positive disease: **post-mastectomy** radiation to the chest wall **and nodal regions**.
- Left-sided disease: the heart sits in the field, so **deep-inspiration breath-hold (DIBH)** is commonly used — the patient holds a deep breath, the lung inflates, and the heart drops away from the chest wall, lowering cardiac dose.

> **Key Point:** DIBH exists because of one OAR — the heart. Anatomy that puts a critical structure near the target is often what justifies a special technique. (This is exactly the breath-hold coaching skill simulated in DIBH practice tools.)

Gastrointestinal (GI)

Anatomy. A long tube — esophagus, stomach, pancreas, rectum — threaded among radiosensitive neighbors. The **small bowel** is the recurring villain: it tolerates limited dose and limits how aggressive pelvic and abdominal plans can be.

Common histology. Mostly **adenocarcinoma**, except the esophagus, which can be SCC (upper) or adenocarcinoma (lower, near the stomach junction).

Routes of spread. Local invasion through the bowel wall, lymphatic to regional nodes, hematogenous to liver (the gut drains to the liver via the portal vein).

Commonly used regimens.

GI site	Typical intent	Commonly used regimen
Esophagus	Definitive or preop, with chemo	~50.4 Gy + chemotherapy
Gastric (stomach)	Adjuvant (after surgery)	~45 Gy
Pancreas	Definitive/adjuvant, with chemo	~50.4 Gy
Rectum	Neoadjuvant (before surgery)	Long-course 50.4 Gy **or** short-course 25 Gy in 5

All of the above are commonly used patterns reflected in NCCN guidance [2]. For **rectal cancer**, the **small bowel is dose-limiting**, which is why patients are often treated prone with a belly board to push bowel out of the field.

> **Common mix-up:** Rectal "short-course" (25 Gy in 5) is not palliation — it is an intense, curative-intent neoadjuvant schedule. Few fractions does not always mean "giving up." Always check the intent.

Genitourinary (GU)

Anatomy. The prostate sits just in front of the rectum and below the bladder — two OARs that flank it like bookends. The kidneys and bladder round out this region.

Common histology. Prostate and bladder cancers are usually **adenocarcinoma** and **urothelial carcinoma** respectively; **renal cell carcinoma (RCC)** arises in the kidney.

Staging and grade notes. Prostate cancer uses the **Gleason score / Grade Group** system, which grades the two most common glandular patterns and sums them — a higher Gleason score means more aggressive disease. Risk grouping (low/intermediate/

high) drives whether **androgen-deprivation therapy (ADT)** — hormone therapy that lowers testosterone — is added [2].

Commonly used regimens.

- Prostate, definitive: a commonly used regimen is **~78–80 Gy** conventionally fractionated, or **hypofractionation/SBRT** in fewer, larger fractions [2][3]. **Gold fiducial markers** are commonly implanted so daily image guidance can localize the moving prostate, and **ADT** is added for higher-risk disease. Rectum and bladder are the key OARs.
- Bladder: **tri-modality bladder preservation** — maximal transurethral resection, then concurrent chemoradiation — as an organ-sparing alternative to cystectomy.
- Renal cell carcinoma: historically **relatively radioresistant** to conventional fractionation; **SBRT** is used for oligometastases (a few isolated metastatic spots) and select primaries.

> **Key Point:** Fiducial markers and image guidance are answers to one question: "how do I hit a target that moves day to day?" The prostate shifts with rectal and bladder filling, so we give it landmarks to track.

Gynecologic (Gyn)

Anatomy. The cervix, uterus (endometrium), and ovaries sit deep in the pelvis. Brachytherapy — placing the radiation source inside or next to the tumor — is central here.

Common histology. Cervix: usually SCC. Endometrium and ovary: usually adenocarcinoma. Cervical cancer is strongly linked to **HPV**.

Commonly used regimens.

- Cervix: a commonly used approach is **~45 Gy to the pelvis** with concurrent **cisplatin**, followed by **brachytherapy**, with dose prescribed to a historic reference called **Point A**; the combined external-beam-plus-brachytherapy total reaches roughly **80–85 Gy EQD2** [2][3]. (EQD2 is the "equivalent dose in 2-Gy fractions" — a way to compare regimens on a common scale.)
- Endometrium: **vaginal-cuff brachytherapy** after hysterectomy, **± pelvic external-beam RT** depending on risk.
- Ovarian: radiation is now **mostly palliative**; systemic chemotherapy and surgery dominate care.

> **Common mix-up:** "45 Gy to the pelvis" sounds modest for a cure — but cervix treatment is a two-part prescription. The brachytherapy boost delivers the high central dose that finishes the job. Never read the external-beam number alone.

Lymphoma

Anatomy and concept. Lymphoma is a cancer of the lymphatic system, so it can appear in nodes anywhere — neck, mediastinum, abdomen. Radiation is usually **consolidation** (mopping up after chemotherapy), not the whole treatment.

Common histology. **Hodgkin lymphoma** (often a mediastinal mass in young adults, with characteristic Reed–Sternberg cells) and **diffuse large B-cell lymphoma (DLBCL)**, the most common aggressive non-Hodgkin type.

Commonly used regimens.

- Hodgkin: **involved-site radiation therapy (ISRT)** consolidation, a commonly used range of **~20–30 Gy** to initially involved sites after chemotherapy [2].
- DLBCL: a commonly used range of **~30–40 Gy** to bulky or residual disease after **R-CHOP** chemotherapy [2].

> **Key Point:** Lymphoma doses are low compared with carcinomas because lymphoma is very radiosensitive. Radiosensitivity, not stinginess, sets these numbers.

Soft-tissue sarcoma

Anatomy and histology. Sarcomas arise from connective tissue — most often in the extremities. They are graded carefully; grade strongly predicts behavior.

Routes of spread. Predominantly **hematogenous to the lung**; nodal spread is **rare**. That is the opposite of most carcinomas and a favorite test pattern.

Commonly used regimens.

- Neoadjuvant (preoperative): a commonly used regimen is **~50 Gy**, then surgery.
- Adjuvant (postoperative): a commonly used range is **~60–66 Gy** [2][3].

Preop uses a smaller field and lower dose (the tumor defines the target); postop uses a higher dose over a larger surgical bed. Both are valid, with different wound-healing and late-effect trade-offs.

Common mix-up: Don't reflexively add elective nodal coverage for sarcoma. Because nodal spread is rare and lung is the real risk, the surveillance focus is the chest, not the nodes.

Pediatric tumors

Guiding principle. A child's tissues are still developing, so radiation late effects (growth, cognition, secondary cancers) weigh heavily. Pediatric radiation is **risk-adapted and de-escalated** — used selectively, at the lowest effective dose, often guided by cooperative-group protocols [3][6].

Common tumors.

- Rhabdomyosarcoma: a soft-tissue sarcoma of childhood; RT is integrated with chemotherapy and surgery.
- Wilms tumor (nephroblastoma): a kidney tumor of young children. A commonly used **flank** dose is **~10.8 Gy** — strikingly low, reflecting both its radiosensitivity and the de-escalation principle.
- Neuroblastoma: a tumor of immature nerve cells; RT is risk-adapted within multimodal therapy.

Key Point: Pediatric doses look "too low" through an adult lens. They aren't undertreatment — they are deliberate de-escalation to protect a growing body.

Skin

Anatomy and histology. Skin cancers are common and superficial, which makes **electron beams** (shallow penetration) and surface techniques ideal. **Basal cell** and **squamous cell carcinoma** are the everyday non-melanoma types; **melanoma** is the aggressive pigment-cell cancer.

Commonly used regimens.

- Basal/squamous cell carcinoma: when RT is the primary treatment, a commonly used range is **~50–70 Gy**, frequently with **electrons** [3].
- Melanoma: radiation is **mostly for metastases or nodal disease** (e.g., after node dissection or for brain/bone mets), not usually the primary curative tool.

Bone metastases (palliation)

Concept. Tumor cells reach bone hematogenously — partly via **Batson's vertebral venous plexus**, a valveless network of spinal veins that lets cells bypass the lungs and seed the spine and pelvis directly. This route helps explain why prostate, breast, and lung cancers so often land in the spine.

Commonly used regimens. For pain palliation, several schedules are commonly used and considered roughly equivalent for pain relief [2][7]:

Schedule	Use case
30 Gy in 10 fractions	More durable control, larger fields
20 Gy in 5 fractions	Middle ground
8 Gy in a single fraction	Convenient; great for poor performance status or short prognosis

Common mix-up: A single 8 Gy fraction is real, effective palliation — not a "weak" treatment. For a patient who struggles to travel, one visit that relieves pain is excellent care.

Summary table: site → intent → commonly used regimen

Site	Typical intent	Commonly used dose/fractionation (example)
Glioblastoma	Definitive (+ temozolomide)	60 Gy / 30
Vestibular schwannoma	Ablative (control)	~12–13 Gy / 1 (SRS)
Brain metastases	SRS or WBRT	SRS, or 30 Gy / 10
Head & neck SCC	Definitive (+ cisplatin)	~70 Gy / 35
Early glottic larynx	Definitive (voice-sparing)	~63–66 Gy
NSCLC (locally advanced)	Definitive (+ chemo)	~60 Gy
NSCLC (early peripheral)	Ablative	50 Gy / 5 (SBRT)
SCLC (limited)	Definitive + PCI	ChemoRT; PCI ~25 Gy / 10
Breast (whole-breast)	Adjuvant	50 Gy / 25 or 40–42.5 Gy / 15 (+ boost)
Esophagus	Definitive/preop (+ chemo)	~50.4 Gy
Gastric	Adjuvant	~45 Gy
Pancreas	Definitive/adjuvant	~50.4 Gy
Rectum	Neoadjuvant	50.4 Gy or 25 Gy / 5
Prostate	Definitive	~78–80 Gy, or hypofx/SBRT
Bladder	Bladder preservation	ChemoRT (tri-modality)
Cervix	Definitive (+ cisplatin + brachy)	~45 Gy + brachy (~80–85 Gy EQD2)
Endometrium	Adjuvant	Vaginal-cuff brachy ± pelvic RT
Hodgkin lymphoma	Consolidation	~20–30 Gy (ISRT)
DLBCL	Consolidation	~30–40 Gy
Soft-tissue sarcoma	Neoadjuvant/adjuvant	~50 Gy / ~60–66 Gy
Wilms tumor	Adjuvant (risk-adapted)	Flank ~10.8 Gy
Skin (BCC/SCC)	Definitive (electrons)	~50–70 Gy
Bone metastases	Palliation	30 Gy / 10, 20 Gy / 5, or 8 Gy / 1

All regimens above are commonly used examples per NCCN and standard references [2] [3]; actual prescriptions vary by protocol and physician.

Check yourself

1. A patient with HPV-positive oropharyngeal SCC and one with HPV-negative disease have identical-looking tumors and nodes. Why might their stage groups differ? Because AJCC 8th edition gives HPV-positive oropharyngeal cancer its own staging table, reflecting its better prognosis [1]. The same anatomic findings map to a different — usually earlier — stage group when HPV-positive.

2. Why is deep-inspiration breath-hold used so often for left-sided breast cancer but rarely for right-sided? The heart sits on the left. Holding a deep breath inflates the lung and pushes the heart away from the chest wall, lowering cardiac dose. The right breast has no such adjacent critical structure, so the technique offers little benefit there.

3. A sarcoma patient asks whether their neck and groin nodes will be treated. How do you frame the answer? Soft-tissue sarcoma spreads hematogenously, classically to the lung, and only rarely to lymph nodes. Elective nodal irradiation is generally not used; surveillance focuses on the chest rather than the nodes.

4. Cervical cancer is treated to "only" about 45 Gy of pelvic external beam, yet it's curative-intent. Explain. Cervix treatment is two-part. External beam covers the pelvis and nodes, and brachytherapy delivers a high central boost (prescribed near Point A). The combined dose reaches roughly 80–85 Gy EQD2, which is what makes it curative — the external-beam number alone understates the total.

5. Why are lymphoma consolidation doses (~20–40 Gy) so much lower than head-and-neck doses (~70 Gy)? Lymphoma is highly radiosensitive, so lower doses control it effectively, especially after chemotherapy. The dose reflects the biology of the tumor, not undertreatment.

6. A frail patient with a single painful spine metastasis lives two hours away. What palliative schedule fits, and why? A single 8 Gy fraction is a strong choice — it provides pain relief comparable to longer schedules for many patients while requiring just one visit, which suits limited performance status and travel difficulty [7]. Spine involvement is common partly because tumor cells reach it via Batson's vertebral venous plexus.

Chapter references

Free, full-text sources

1. NCI — PDQ cancer treatment summaries by site (histology, staging, routes of spread, typical management): https://www.cancer.gov/types.

2. NCCN — Clinical Practice Guidelines in Oncology (free with registration): https://www.nccn.org/guidelines/category_1.

For deeper reading (library / textbook)

1. Gunderson LL, Tepper JE. *Clinical Radiation Oncology*. Elsevier — disease-site chapters and dose/fractionation.
2. Washington CM, Leaver DT. *Principles and Practice of Radiation Therapy*. Elsevier.
3. AJCC *Cancer Staging Manual*, 8th ed. — TNM staging.

Chapter 6 — Treatment Volume Localization

What you'll learn

- How CT simulation captures the patient in the exact position they will be treated in, and why slice thickness, scan range, and contrast matter.
- The difference between a 3D snapshot and a 4D-CT that "watches" the patient breathe.
- How immobilization devices, reference marks, tattoos, and the isocenter turn a plan into a repeatable daily setup.
- The ICRU target volumes — GTV, CTV, ITV, PTV — and exactly what each one adds.
- The main image-guided radiation therapy (IGRT) methods, and how to read an image "match" the way a therapist does at the machine every day.
- What a 6-degrees-of-freedom couch correction means, and when a patient needs to be re-simulated (adaptive radiotherapy).

Why this matters

Radiation only helps if it lands on the tumor and spares the healthy tissue around it. Everything in this chapter — the simulation scan, the foam cradle, the tiny ink tattoo, the daily image match — exists to put the beam in the same correct place every single day. As a therapist, image matching will be one of your most repeated and most consequential tasks, so we will spend extra time making it clear.

1. CT Simulation: Building the Patient's Blueprint

Simulation is the planning appointment where we record the patient's anatomy and define the treatment position. Think of it as the architectural survey before a house is built. Nothing about the rest of the treatment course is repeatable unless this first scan is done carefully.

In modern departments, simulation is done on a **CT simulator** — a CT (computed tomography) scanner with a flat couch top that matches the treatment couch, plus a set of **lasers** mounted on the room walls and ceiling. The single most important rule: **scan the patient in the treatment position.** If they will be treated lying on their back with arms over their head, that is exactly how they are scanned. The CT images become the 3D model the dosimetrist plans on, so the model must match reality.

Slice thickness and scan range

The CT image is built from many thin cross-sectional slices, like a loaf of sliced bread. **Slice thickness** is how thick each slice is.

Setting	Typical slice thickness	Why
Stereotactic (SRS/SBRT, very precise small targets)	1–3 mm	Thinner slices = finer detail and smoother digitally reconstructed images
Conventional treatment	3–5 mm	Good detail with manageable data size

Thinner slices give more detail but make larger image sets. The **scan range** is how much of the body is imaged — always generous enough to include the target, all the organs at risk nearby, and enough length above and below for the planned beams to enter and exit.

Contrast and artifacts

Contrast agents make certain tissues stand out.

- **IV (intravenous) contrast** brightens blood vessels and many tumors, helping define the target. Note: contrast changes the apparent tissue density the planning system reads, so departments have protocols for handling this.
- **Oral contrast** outlines the stomach and bowel so the bowel can be avoided.

Watch for **artifacts** — errors in the image. **Metal** (hip prostheses, dental fillings) causes **beam-hardening artifacts**: bright and dark streaks that distort the picture and the dose calculation. We plan around these, sometimes with special metal-artifact-reduction settings.

> **Key Point:** The CT simulation scan is the reference everything else is measured against. If the patient is positioned poorly at sim, every treatment day inherits that error.

2. 3D vs. 4D-CT: Capturing Motion

A standard CT scan is a **3D** image — a single frozen snapshot. That is fine for anatomy that holds still. But the lungs, liver, and pancreas move as the patient breathes. A snapshot of a moving target is like a photo of a runner: it shows where they were in one instant, not the full path they travel.

4D-CT solves this. It adds time as the fourth dimension. The scanner records the breathing pattern (with a belt or an external marker) while it images, then **sorts** the slices by **breathing phase** — full inhale, mid-breath, full exhale, and stages in between. The result is a movie of the anatomy across the breathing cycle.

Why we care: 4D-CT shows the full range a tumor travels during breathing. That range is exactly what we need to build the **internal margin** (the ITV, below) and to design **motion management** strategies like breath-hold or gating.

Common mix-up: 3D-CT is not "lower quality" than 4D-CT — it is simply a still image. 4D-CT is specifically for moving targets and produces much more data, so it is reserved for sites where breathing motion matters.

3. Immobilization: Holding the Position

A great simulation is useless if the patient lies differently tomorrow. **Immobilization devices** lock the patient into a reproducible position. They are matched to the body region:

- **Thermoplastic mask** — a mesh sheet softened in warm water, molded to the face/head/neck, then hardened. It clips to the couch and holds the head still. Good masks reproduce position to within roughly **3 mm**.
- **Breast board** — an inclined board with arm and hand grips that raises the arms out of the field and sets a consistent torso angle.
- **Vacuum bag (vac-lok / cradle)** — a bag of foam beads molded around the patient; air is pumped out so it stiffens into a custom cradle. Common for body SBRT and pelvis.
- **Headframe / stereotactic frame** — the most rigid option, historically bolted on for cranial SRS; very small setup error.

There is a direct trade: **the tighter and more reproducible the immobilization, the smaller the setup margin we have to add** around the target. Less wobble means we can safely treat a smaller volume, sparing more healthy tissue. This is why a rigid SRS frame allows millimeter margins while a loose setup forces a bigger, more generous margin.

4. Reference Marks, Tattoos, and the Isocenter

Once the patient is positioned and scanned, we mark them so the position can be found again.

- **Reference marks / tattoos** — tiny permanent **India-ink dots** placed at reference points on the skin (often three: one anterior, two lateral). They are aligned to the room lasers and give the therapist a starting point each day. They are deliberately small and permanent so they don't fade and don't bother the patient long-term.

- **The isocenter** is the heart of the whole geometry. It is the single point in space where the rotation axes of the **gantry** (the rotating beam head), the **collimator** (the beam-shaping head), and the **couch** all intersect. Imagine three spinning hoops all threaded through one bead — that bead is the isocenter. The beam can rotate all the way around the patient and always passes through this one point.

We place the isocenter inside or near the target at planning, then mark its location so it can be reproduced. Two things make that possible:

1. The room's **three-plane lasers** (sagittal, coronal, axial) project lines that cross at a fixed point in the room.
2. We **record the isocenter's CT coordinates** relative to the reference tattoos — the measured shift (in cm) from the tattoo point to the true isocenter.

Each treatment day, the therapist lines the tattoos up to the lasers, then applies the recorded shift to drive the couch to the isocenter. After that, imaging fine-tunes it.

> **Key Point:** The isocenter is a point, not a region — the intersection of the gantry, collimator, and couch rotation axes. Lasers reproduce it in the room; recorded CT coordinates tell us where it sits relative to the patient's marks.

5. ICRU Target Volumes (Reports 50, 62, and 83)

The **ICRU** (International Commission on Radiation Units and Measurements) defined a standard set of nested volumes so that everyone — physician, dosimetrist, therapist — means the same thing by the same words [1][2][3]. Picture a set of Russian nesting dolls: each volume contains the one before it and adds one specific kind of uncertainty.

Volume	Stands for	What it adds	Plain-language meaning
GTV	Gross Tumor Volume	Nothing — it is the starting point	The tumor we can actually **see or feel** (on imaging or exam). **No margin.**
CTV	Clinical Target Volume	Margin for **microscopic spread** + elective nodes	GTV plus a rim where disease likely exists but is invisible, plus nearby lymph node regions we treat electively.
ITV	Internal Target Volume	Margin for **internal motion**	CTV plus the range the target moves with breathing/filling — built from **4D-CT**.
PTV	Planning Target Volume	Margin for **setup uncertainty**	CTV/ITV plus a buffer for day-to-day positioning error. The volume we actually aim the dose at.
OAR	Organ At Risk	—	A healthy structure we must protect (cord, lung, rectum).
PRV	Planning organ at Risk Volume	A safety margin **around the OAR**	The OAR plus a margin, so motion/setup error doesn't accidentally overdose it.

A few accuracy rules worth memorizing:

- **GTV has no margin.** It is purely what is visible or palpable.

- **If the tumor was surgically removed (resected), the GTV = 0** — there is nothing gross left to see. We still treat a CTV (the tumor bed and at-risk tissue).
- **CTV adds microscopic disease** (the invisible spread) and elective nodes.
- **ITV adds internal motion** only.
- **PTV adds setup uncertainty** only.

The van Herk margin recipe

How big should the PTV margin be? A common, evidence-based answer is the **van Herk margin recipe** [4], often written:

> margin ≈ $2.5\,\Sigma + 0.7\,\sigma$

where **Σ (sigma)** represents **systematic** errors (the same error every day, like a consistent sim or planning offset) and **σ** represents **random** errors (day-to-day variation). Systematic errors are weighted more heavily because they shift every fraction in the same direction. You don't need to compute it by hand for the registry, but understand the idea: **the margin is a calculated cushion, and better immobilization plus daily imaging shrinks it.**

> **Common mix-up:** Students often swap ITV and PTV. Remember the order of the uncertainties: **ITV = motion inside the body; PTV = error in setting the body up.** ITV is about the tumor moving; PTV is about the patient being positioned slightly differently each day.

6. Image-Guided Radiation Therapy (IGRT)

Tattoos and lasers get the patient close. **IGRT** — taking an image at the machine and comparing it to the plan — gets them exactly right. This section is the heart of the chapter, so we will go slowly.

The big idea: matching a moving image to a reference image

Every form of IGRT does the same fundamental thing:

- The **reference image** is from the planning CT — where the anatomy is supposed to be. (In our trainer's color scheme, the reference is shown in **orange**.)
- The **moving image** is taken today at the machine — where the anatomy actually is right now. (Shown in **blue**.)

The therapist's job is to **slide and rotate the today image until it lines up with the plan image.** When you nudge the images on screen, you are really telling the system how far off the patient is. The system then converts that into a **couch correction** — physically moving the treatment couch so the patient's anatomy sits exactly where the plan expects. Match on screen, then the couch moves the patient to agree with the plan.

Think of it like aligning two printed transparencies on an overhead projector: you shift and twist the top sheet until its lines fall exactly on the bottom sheet's lines. The amount you shifted is your correction.

What a 6DOF couch correction means

A couch can correct position in up to **six degrees of freedom (6DOF)** — six independent ways to move:

Three translations (sliding):

Axis	Direction	Memory aid
Lat (lateral)	left ↔ right	side to side
Long (longitudinal)	head ↔ feet	up and down the table
Vert (vertical)	up ↔ down	raising/lowering the couch

Three rotations (tilting/twisting):

Rotation	Motion	Everyday image
Pitch	nodding	like nodding "yes"
Roll	side tilt	like tipping your head toward a shoulder
Yaw	turning	like shaking "no"

A couch that only does the three translations is a **3DOF** (or 4DOF, adding yaw) couch. A full **6DOF couch** can also correct the three rotations, which matters most for high-precision cases like cranial SRS and spine SBRT, where a tiny tilt throws the target off.

> **Key Point:** A couch shift is just the real-world result of your on-screen match. You align the today image to the plan image; the couch then moves the patient by exactly that amount so the beam hits the planned spot.

Now the specific modalities.

6.1 2D/2D Planar kV Imaging

The machine takes two **planar** (flat, X-ray-style) **kV** (kilovoltage — diagnostic-energy) images at right angles to each other: typically an **AP** (anterior–posterior, front-to-back) view and a **lateral** (side) view. This pair is called an **orthogonal** pair because the two views are 90° apart, like a front photo and a side photo of a building.

Each live image is matched to a **DRR (digitally reconstructed radiograph)** — a synthetic X-ray the planning system creates by ray-tracing through the planning CT. So you are comparing today's real X-ray to a "predicted" X-ray from the plan. You align by either:

- **Bone match** — line up the skeleton (vertebrae, pelvic bones), used when the target tracks with bone.
- **Fiducial match** — line up implanted **fiducial markers** (small gold seeds placed in or near the tumor), used when the target moves relative to bone (e.g., prostate).

The weakness: planar kV images have **poor soft-tissue contrast.** You can see bone and metal seeds well, but the tumor and soft organs are faint or invisible — which is exactly why fiducials are implanted for sites like the prostate.

6.2 MV Portal Imaging / EPID

MV (megavoltage) portal imaging uses the **treatment beam itself** to make the image, captured by an **EPID (electronic portal imaging device)** — a flat panel behind the patient. Because the treatment beam is so high-energy, the images have **low contrast** (everything looks washed out and gray) and it delivers a little extra dose. It is the **older** method, largely replaced by kV imaging and CBCT, but you should recognize the term: portal imaging = using the MV treatment beam to verify the field.

6.3 kV Cone-Beam CT (CBCT)

CBCT (cone-beam CT) is the big upgrade. Instead of two flat images, the kV source and detector rotate around the patient to acquire a **full 3D volumetric image** right on the treatment machine [6]. This unlocks two major advantages:

- **Soft-tissue matching** — because it is a true 3D image, you can match on the actual soft-tissue target and nearby organs (prostate, bladder, rectum), not just bone.
- **Full 6DOF correction** — you view the patient in three planes (**axial, coronal, sagittal**) and can correct all three translations and all three rotations.

In a CBCT match you scroll through slices and fuse the planning CT (orange) with today's CBCT (blue), shifting and rotating until soft-tissue landmarks overlap. This is the modern workhorse of IGRT.

> **Common mix-up:** "kV" vs "MV" describes the **energy** of the imaging beam (diagnostic vs treatment). "2D vs CBCT" describes the **dimension** of the image (flat pair vs full volume). A kV beam can make either a 2D planar image or a 3D cone-beam CT.

6.4 Implanted Fiducial Markers and Transponders

When the target is invisible on imaging or moves relative to bone, we implant markers:

- **Gold seeds** — small inert gold fiducials placed in the **prostate, pancreas, lung, or liver.** They show up crisply on kV images, giving a reliable target to match even when the surrounding tissue is faint.
- **Electromagnetic transponders (Calypso)** — tiny implanted "beacons" that broadcast their position continuously to an antenna array. This allows **real-time tracking** of the target during treatment with no imaging dose, often described as "GPS for the body."

6.5 Surface-Guided RT (SGRT)

SGRT (surface-guided radiation therapy) uses **optical cameras** to map the **skin surface** in 3D, comparing it to a reference surface from the plan. Because it uses light, not X-rays, it adds **no extra radiation dose** and can run continuously. Its three main jobs:

1. **Setup** — position the patient by matching the skin surface, often reducing reliance on tattoos.
2. **Intrafraction monitoring** — watch the surface during the beam and pause if the patient drifts.
3. **DIBH (deep-inspiration breath-hold)** — coach the patient to hold a deep breath. In left-breast treatment, a deep breath pushes the heart away from the chest wall, sparing it. SGRT shows whether the patient is holding within the correct **gating window**, and the beam only turns on while they are in that window.

7. Adaptive Radiotherapy and Re-Simulation

A treatment course runs over many weeks, and the patient's anatomy can change. When it changes enough that the original plan no longer fits, we adapt.

Common triggers:

- **Weight loss** — the mask or immobilization no longer fits snugly; the external contour has shifted.
- **Tumor shrinkage** — a head-and-neck or lung tumor that responds well can shrink away from the planned volume, meaning the plan now treats too much normal tissue (or the OARs have moved into the dose).
- **Organ filling changes** — a fuller or emptier **bladder or rectum** shifts the prostate and changes the dose to nearby organs.

Adaptive radiotherapy (ART) means modifying the plan to match the patient's current anatomy. This may be a quick **online adaptation** (re-planning at the machine using today's CBCT) or a full **re-simulation** — bringing the patient back for a new CT sim and a brand-new plan. The principle is simple: the plan should always match the patient in front of you, not the patient from week one.

> **Key Point:** Daily imaging fixes where the patient is today. Adaptive radiotherapy fixes the situation where the patient's anatomy itself has changed and the original plan no longer fits.

Check yourself

1. **A patient's lung tumor moves with breathing. Which volume captures that motion, and what imaging is used to build it?** The ITV (Internal Target Volume) captures breathing motion, and it is built from a 4D-CT, which sorts images by breathing phase to show the full range of motion.

2. **A tumor was completely surgically removed before radiation. What is the GTV, and why?** The GTV is 0 (zero). GTV is only the visible or palpable gross tumor; if it has been resected, there is nothing gross left to outline. A CTV is still treated to cover the tumor bed and microscopic disease.

3. **Put these in order from innermost to outermost and state what each adds: PTV, GTV, CTV, ITV.** GTV (visible tumor, no margin) → CTV (adds microscopic spread and elective nodes) → ITV (adds internal motion) → PTV (adds setup uncertainty).

4. **Why are gold fiducial markers implanted in the prostate for 2D/2D kV imaging?** Because planar kV images have poor soft-tissue contrast, the prostate itself is hard to see. The gold seeds are clearly visible, giving a reliable target to match — and they track the prostate's motion, which moves relative to bone.

5. **In plain terms, what does "matching the moving image to the reference image" produce, and what physically happens next?** The therapist aligns today's at-machine image (moving) to the planning-CT image (reference). The misalignment is converted into a couch correction (shift), and the treatment couch physically moves the patient so their anatomy matches the plan.

6. **What does a 6DOF couch correct that a 3DOF couch cannot, and when does it matter most?** A 6DOF couch corrects the three rotations — pitch, roll, and yaw — in addition to the three translations (Lat, Long, Vert). It matters most for high-precision cases like cranial SRS and spine SBRT, where a small tilt would miss the target.

Chapter references

Free, full-text sources

1. IAEA — *Radiation Oncology Physics: A Handbook for Teachers and Students* (free PDF; simulation, treatment planning, target-volume and image-guidance chapters): https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1196_web.pdf.
2. AAPM Reports — respiratory motion (TG-76), CBCT/IGRT QA (TG-179), IGRT (TG-104/TG-147), all free: https://www.aapm.org/pubs/reports/.

For deeper reading (library / standards)

1. ICRU Reports 50, 62, and 83 — the defining documents for GTV/CTV/ITV/PTV/PRV and margins (ICRU; purchase).
2. Washington CM, Leaver DT. *Principles and Practice of Radiation Therapy*. Elsevier — simulation and IGRT chapters.

This chapter offers original educational explanations. It is not affiliated with or endorsed by the ARRT and does not reproduce actual exam questions.

Chapter 7 — Prescription & Dose Calculation

What you'll learn

- What a radiation prescription actually specifies, and where the dose is "prescribed to."
- The single most important setup distinction in dose calculation: **SSD versus SAD**, and which depth-dose table each one uses.
- How to turn a rectangular field into an **equivalent square**, and why we bother.
- What **PDD, TMR, TPR, and TAR** mean — in plain words — and how to read a dose off them.
- How to do a **monitor-unit (MU) calculation** step by step, for both setups, including wedges and trays.
- The special cases that trip people up: tissue inhomogeneity, field-junction gaps, and electron-beam ranges.
- How we judge a finished plan with the homogeneity index, conformity index, and the dose-volume histogram.

If math makes you tense, take a breath — this chapter is built for you. Every formula is introduced in words first, then shown with a fully worked example where we write out every step. You will never be asked to do anything more complicated than multiply, divide, and square a number. Go slowly, keep a calculator handy, and let each example sink in before moving on.

Part 1 · The prescription and the language of dose

A radiation **prescription** is the physician's order. At minimum it names the **total dose**, the **dose per fraction**, the **number of fractions**, and the **technique and energy**. For example, "60 Gy in 30 fractions, 6 MV, IMRT" means 2 Gy each day for 30 days.

Dose has to be prescribed **to a point or volume**, because dose is not the same everywhere in the beam. Historically we prescribed to the **ICRU reference point** — a clearly defined point (often at or near the isocenter, in the center of the target) chosen so that different clinics report dose the same way. Modern conformal and intensity-modulated plans usually prescribe so that a chosen **isodose line** covers the target, but the ICRU reference point is still how we talk about and report the dose.

> **Key Point:** "Dose" always comes with a location. A prescription of 200 cGy means 200 cGy at the prescription point or isodose line — not everywhere in the beam.

The physician also picks a **beam energy** based on how deep the target is (Chapter 3: higher energy = deeper d_max and more penetration) and may add **beam modifiers** such as **wedges** to shape the dose. All of these choices feed into the calculation that tells the machine how long to stay on — the monitor-unit calculation we build toward in this chapter.

Part 2 · Two ways to set up: SSD versus SAD

Before any numbers, you must know **how the patient is set up**, because it decides which table you use. There are two setups, and confusing them is the number-one dose-calc mistake.

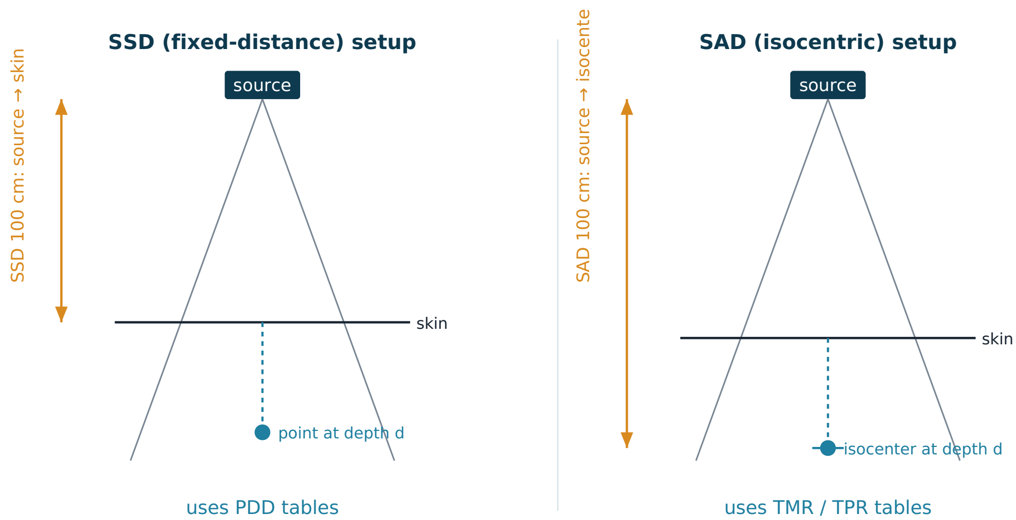

Figure 7.1 — In an SSD setup the fixed 100 cm runs from the source to the patient's skin, and the target sits at some depth below it. In an SAD setup the fixed 100 cm runs all the way to the isocenter inside the patient. The difference decides which depth-dose table you use.

- **SSD (Source-to-Surface Distance) setup.** You fix the distance from the source to the **skin** (commonly 100 cm). The target is then at some depth below that fixed point. SSD setups use **PDD** tables.
- **SAD (Source-to-Axis Distance) setup**, also called **isocentric**. You fix the distance from the source to the **isocenter**, which sits inside the patient at the target. The skin is somewhere above it. SAD setups use **TMR** (or TPR) tables. This is the standard for modern linac treatments because you can rotate the gantry all the way around without moving the patient.

> **Common mix-up:** In **SSD** the fixed distance lands on the **skin**; in **SAD** the fixed distance lands on the **isocenter inside the patient**. SSD → PDD. SAD → TMR/TPR. Lock this pairing in and half the chapter falls into place.

Part 3 · Field size and the equivalent square

The beam's **field size** (set by the jaws and the multileaf collimator) affects how much scattered dose reaches a point — a bigger field means more scatter and a bit more dose. Our depth-dose tables are written for **square** fields, but real fields are often **rectangular**. So we convert a rectangle into the square that behaves the same way: its **equivalent square**.

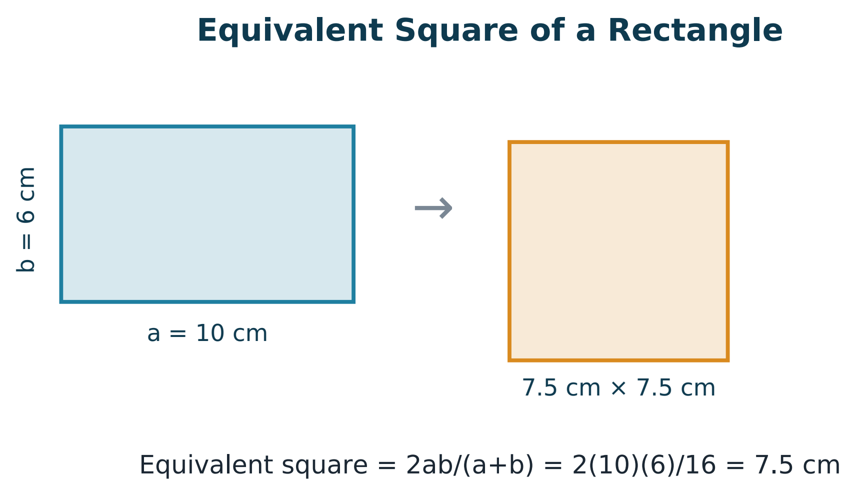

Figure 7.2 — A 10 cm × 6 cm rectangle scatters dose like a 7.5 cm × 7.5 cm square. We use the equivalent-square size to look up depth-dose and scatter factors.

> **Key Point: Equivalent square = 2ab / (a + b)** (the same as 4 × Area ÷ Perimeter), where a and b are the rectangle's sides.

Worked example. A field is 10 cm × 6 cm. What is its equivalent square?

```
Equivalent square = 2ab / (a + b)
                  = 2 × 10 × 6 / (10 + 6)
                  = 120 / 16
                  = 7.5 cm
```

So you would look up your depth-dose and scatter numbers for a **7.5 cm × 7.5 cm** field.

Part 4 · Depth-dose functions

These four "ratios" all describe how dose changes with depth. You do not have to derive them — you just need to know what each one means and which setup it belongs to.

- **PDD (Percent Depth Dose).** The dose at a depth, written as a percentage of the dose at d_max:

> **Key Point: PDD = (dose at depth d ÷ dose at d_max) × 100.** PDD belongs to **SSD** setups. PDD goes up with higher energy, larger field size, and longer SSD; it goes down with depth.

Worked example. A beam deposits 100 cGy at d_max. If the PDD at 10 cm is 66%, what is the dose at 10 cm depth?

```
Dose at depth = dose at d_max × (PDD ÷ 100)
              = 100 cGy × (66 ÷ 100)
              = 66 cGy
```

- **TMR (Tissue-Maximum Ratio).** The dose at a depth divided by the dose at d_max **at the same point in space** — so it does not change when you move the source closer or farther. That distance-independence is exactly what an isocentric (**SAD**) setup needs.
- **TPR (Tissue-Phantom Ratio).** Almost the same as TMR, but the reference depth is a fixed depth (often 10 cm) instead of d_max. Also used for SAD.
- **TAR (Tissue-Air Ratio).** An older ratio (dose in tissue ÷ dose in air at the same point). You should recognize the name, but TMR has largely replaced it for high-energy beams.

> **Common mix-up: PDD changes with distance** (it is tied to a fixed SSD), while **TMR/TPR do not** (they are tied to a fixed point). That is the whole reason isocentric treatments use TMR.

Part 5 · The inverse-square law, revisited

Recall from Chapter 3 (Figure 3.3) that dose rate falls with the **square** of the distance from the source: $\mathbf{D_2 = D_1 \times (d_1/d_2)^2}$. In dose calculation this is how you correct for treating at a distance other than the calibration distance.

Worked example. A machine gives 200 cGy at 100 cm. What is the dose rate at 120 cm?

```
D2 = D1 × (d1 / d2)²
   = 200 × (100 / 120)²
   = 200 × (0.833)²
   = 200 × 0.694
   ≈ 139 cGy
```

Part 6 · Scatter, wedges, and trays — the correction factors

A real calculation includes a handful of multiplying factors. Each one answers "how much does this change the dose per MU?"

- **Collimator (head) scatter, S_c.** Extra dose scattered from the machine head as the field opens up. It depends on the **collimator setting**, measured in air.
- **Phantom (patient) scatter, S_p.** Extra dose scattered from the **patient's own tissue**. It depends on the field size at the patient. Together, **S_cp = S_c × S_p** is the total scatter factor.

> **Common mix-up:** **S_c is scatter from the machine** (collimator/head, measured in air); **S_p is scatter from the patient** (phantom). Both grow as the field gets bigger.

- **Wedge factor, WF.** A **wedge** is a metal block, thick on one side, that sits in the beam to tilt the dose. Because it absorbs part of the beam, it lowers the dose per MU — so you need more MUs. Its effect is captured by the wedge factor (a number less than 1, e.g., 0.70 for a steep wedge).

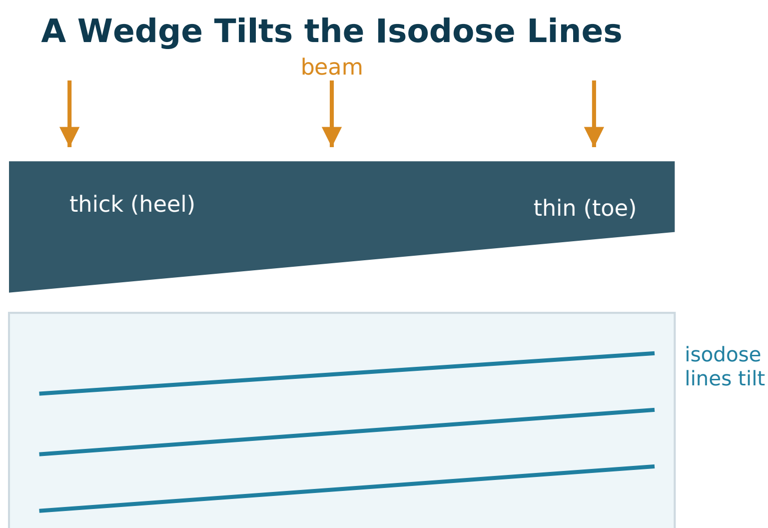

Figure 7.3 — A wedge is thick on one edge (the heel) and thin on the other (the toe). The thick heel absorbs more of the beam, so the isodose lines tilt — useful for matching sloping surfaces or blending adjacent fields.

- **Tray (block) factor, TF.** A tray that holds shielding blocks also absorbs a little beam (often around 0.97), so it too is a factor in the calculation.

Part 7 · Putting it together: the monitor-unit calculation

A **monitor unit (MU)** is how a linac measures its output — roughly, "how long the beam stays on." The MU calculation asks: given everything above, how many MUs deliver the prescribed dose? The logic is always the same — **divide the dose you want by the dose you get per MU**, after accounting for every factor.

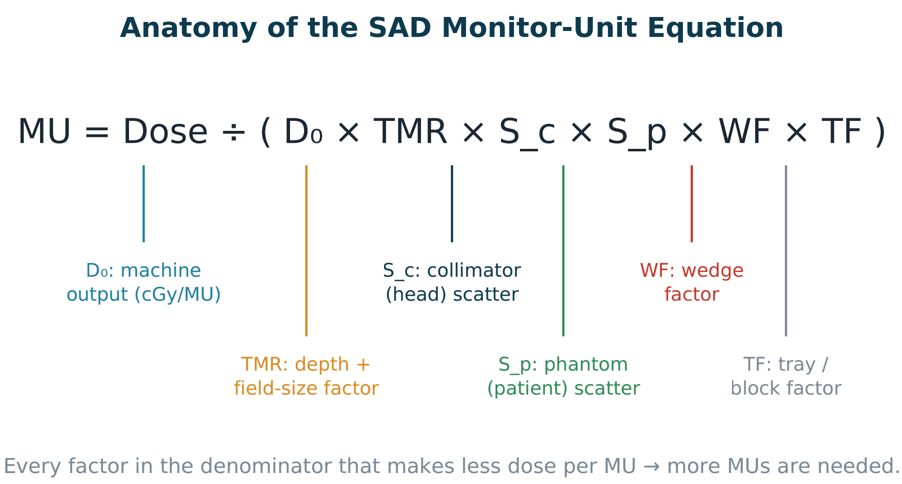

Figure 7.4 — Reading the SAD monitor-unit equation factor by factor. Anything in the denominator that lowers the dose per MU (a wedge, depth, a tray) means more monitor units are needed to reach the prescribed dose.

For an **isocentric (SAD)** setup:

```
MU = Dose ÷ ( D0 × TMR × Sc × Sp × WF × TF )
   D0  = machine output (cGy per MU) at the reference conditions
   TMR = tissue-maximum ratio at the treatment depth & field size
   Sc, Sp = collimator and phantom scatter factors
   WF, TF = wedge and tray factors
```

Worked example (SAD, open field). Prescribe **200 cGy** to the isocenter. The machine is calibrated to **1.000 cGy/MU**. At this depth and field size, **TMR = 0.85**; **S_c = 1.00**, **S_p = 1.00**; no wedge or tray (WF = TF = 1.00).

```
MU = 200 ÷ (1.000 × 0.85 × 1.00 × 1.00 × 1.00 × 1.00)
   = 200 ÷ 0.85
   ≈ 235 MU
```

Now add a wedge. Keep everything the same but insert a wedge with **WF = 0.70**.

```
MU = 200 ÷ (1.000 × 0.85 × 0.70)
   = 200 ÷ 0.595
   ≈ 336 MU
```

See how adding the wedge — which lowers dose per MU — raises the MUs from 235 to 336. Every factor below 1 in the denominator pushes the MU count up.

For an **SSD** setup, the only change is that **PDD/100 replaces TMR**:

```
MU = Dose ÷ ( D0 × (PDD/100) × Sc × Sp × WF × TF )
```

Worked example (SSD). Deliver **200 cGy** to a depth where **PDD = 66%**, with D0 = 1.000 cGy/MU and all other factors = 1.00.

```
MU = 200 ÷ (1.000 × 0.66)
   ≈ 303 MU
```

For **electron beams**, the dose is normalized at **d_max** rather than at depth, and electron-specific output factors are used — but the same "dose ÷ dose-per-MU" logic applies.

Part 8 · Corrections and special cases

Tissue inhomogeneity

Our basic tables assume the body is all water. Real patients contain **lung** (low density — the beam passes through more easily, so dose downstream is higher than the water assumption) and **bone** (high density — more attenuation). Treatment-planning systems correct for this. Older hand methods include the **effective-depth** method and the **Batho power-law** method; modern systems use convolution or Monte Carlo algorithms that handle it automatically. The takeaway for you: **ignoring lung and bone can be off by 5–20%**, which is why these corrections matter most in the chest.

Field-junction gaps

When two beams sit next to each other (for example, treating the spine in two segments), their edges **diverge** outward as they go deeper. If you butt them right together at the skin, they will **overlap** at depth and create a hot spot. The fix is to leave a small **gap at the skin** so the field edges meet cleanly at the target depth.

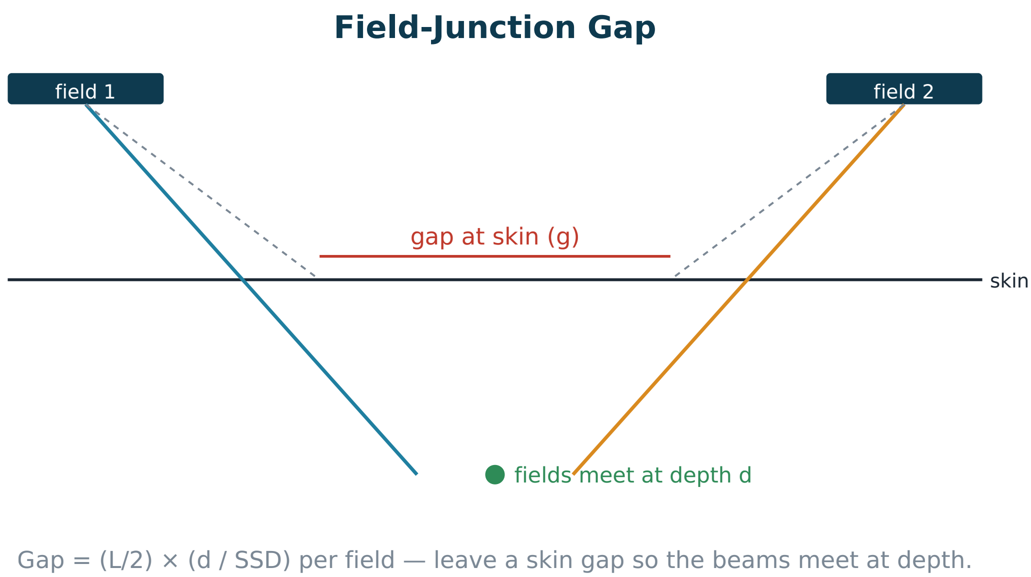

Figure 7.5 — Each beam spreads as it goes deeper. Leaving a calculated gap at the skin lets the diverging edges meet at the treatment depth instead of overlapping (a hot spot) or separating (a cold spot).

> **Key Point: Gap = (L/2) × (d / SSD)** for each field, where L is the field length, d is the junction depth, and SSD is the source-to-surface distance. Add the two fields' contributions for the total skin gap.

Worked example. Two fields, each **20 cm** long, are to join at a depth of **8 cm**, with **SSD = 100 cm**.

```
Each field's contribution = (L / 2) × (d / SSD)
                          = (20 / 2) × (8 / 100)
                          = 10 × 0.08
                          = 0.8 cm
Total gap = 0.8 cm + 0.8 cm = 1.6 cm
```

Electron beams: ranges and energy choice

Electrons are wonderful for **superficial** targets because they deposit dose and then **stop**, sparing whatever is behind them. Three "range" numbers describe an electron beam's depth-dose curve:

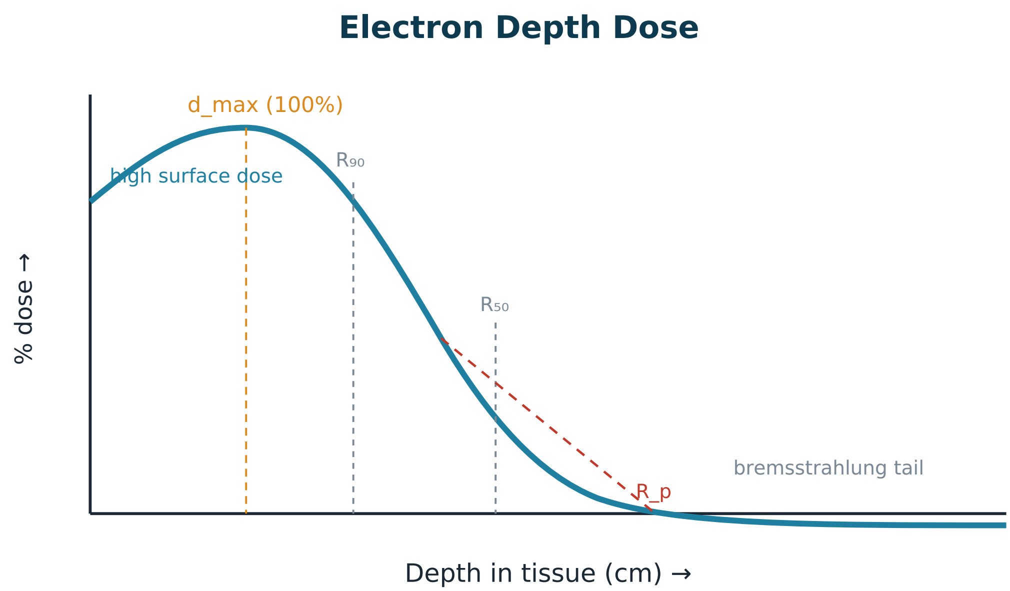

Figure 7.6 — Electrons give a high surface dose, a quick d_max, then a steep fall-off and a small bremsstrahlung tail. R_{90} (the therapeutic range) is the depth that still gets 90% of the dose; R_{50} is the half-dose depth; Rp is the practical range where the steep slope, extended, hits the axis.

- **R_{90}** — the depth still receiving 90% of the dose. This is the **therapeutic range**: cover your target inside R_{90}.
- **R_{50}** — the depth of 50% dose (used to define the beam's quality/energy).
- **Rp** — the **practical range**, where the steepest part of the curve, extended down, crosses the axis.

A handy rule of thumb is that **$R_{90} \approx E \div 3.2$** (in cm, for energy E in MeV), and the practical range **$Rp \approx E \div 2$**. Both depths get **deeper as energy rises**.

Worked example. A lesion needs coverage to **3 cm** deep. Roughly what electron energy do you need?

```
We need R90 ≥ 3 cm.
R90 ≈ E / 3.2   →   E ≈ 3.2 × R90 = 3.2 × 3 = ~10 MeV
Pick the next available energy up (e.g., 12 MeV) to be safe.
```

Part 9 · Judging the plan: HI, CI, and the DVH

Once a plan exists, we score it with a few quick numbers:

- **Homogeneity Index (HI)** — how uniform the dose is across the target. A common form is HI = $D_5 \div D_{95}$ (the dose to the hottest 5% over the dose to the coldest 95%); closer to **1.0** is more uniform.

- **Conformity Index (CI)** — how tightly the prescription dose wraps the target: the volume covered by the prescription isodose ÷ the target volume. Around **1.0** is ideal (covers the target without spilling dose into normal tissue).
- **Dose-Volume Histogram (DVH)** — the single most useful plan-evaluation picture.

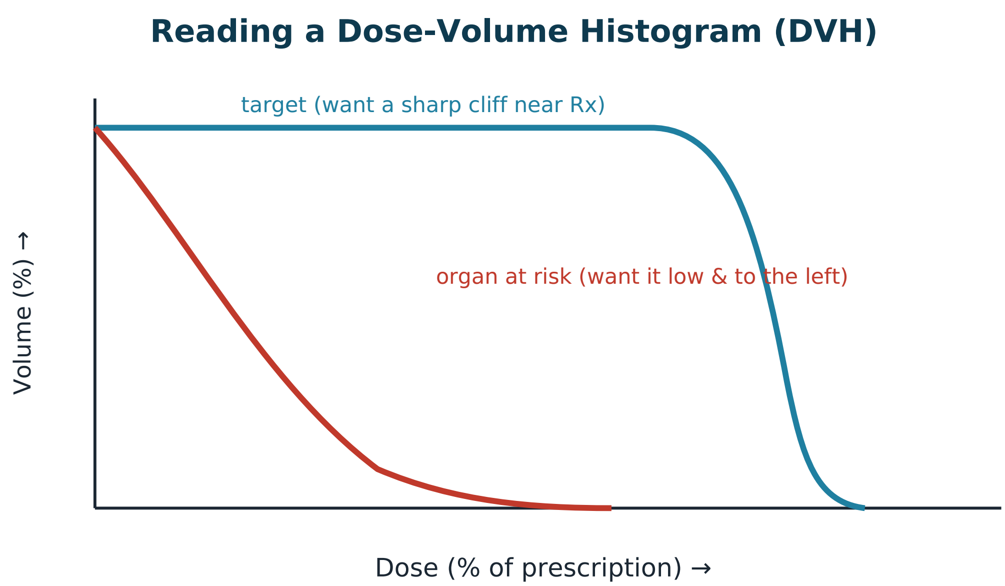

Figure 7.7 — A DVH plots, for each structure, the volume receiving at least a given dose. You want the target curve to stay high and then drop in a sharp cliff near the prescription dose, and each organ-at-risk curve to fall away early (low and to the left).

> **Key Point:** On a DVH, a **good target** curve holds near 100% volume and then falls off a cliff right at the prescription dose; a **well-spared organ at risk** drops off early, staying low and to the left.

Check yourself

1. Which depth-dose table goes with an SSD setup, and which goes with a SAD setup? SSD uses PDD; SAD (isocentric) uses TMR (or TPR).

2. Find the equivalent square of a 15 cm × 5 cm field. 2ab/(a+b) = 2 × 15 × 5 / (15 + 5) = 150 / 20 = 7.5 cm.

3. A beam gives 100 cGy at d_max. If the PDD at 12 cm is 60%, what is the dose at 12 cm? 100 × (60/100) = 60 cGy.

4. Calculate the MU for 250 cGy to the isocenter with D0 = 1.000 cGy/MU, TMR = 0.80, and all other factors = 1.00. MU = 250 ÷ (1.000 × 0.80) = 312.5 ≈ 313 MU.

5. You insert a wedge (WF = 0.75) into the setup in question 4. What happens to the MU, and what is the new value? The MU goes up: MU = 250 ÷ (0.80 × 0.75) = 250 ÷ 0.60 ≈ 417 MU.

6. Two 18 cm fields join at 6 cm depth, SSD = 100 cm. What is the total skin gap? Each field: (18/2) × (6/100) = 9 × 0.06 = 0.54 cm. Total = 0.54 + 0.54 = 1.08 cm.

7. Why do we use electron beams for superficial lesions? Electrons deposit their dose over a short range and then stop, sparing the deeper tissue behind the target.

8. On a DVH, what does an ideal target curve look like? It stays near 100% volume across the lower doses and then drops in a sharp cliff right around the prescription dose — meaning the whole target gets the dose and little more.

Chapter references

Free, full-text sources

1. AAPM — Task Group 71, Monitor unit calculations for external photon and electron beams (free report; equivalent square, S_c/S_p, SSD & SAD MU, electron MU): https://www.aapm.org/pubs/reports/detail.asp?docid=151.
2. IAEA — Radiation Oncology Physics: A Handbook for Teachers and Students (free PDF; PDD/TMR/TPR, dose calculation, electron ranges, treatment planning): https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1196_web.pdf.
3. AAPM Reports — tissue inhomogeneity corrections (Report 85 / TG-65), free: https://www.aapm.org/pubs/reports/.

For deeper reading (library / textbook)

1. Khan FM, Gibbons JP. The Physics of Radiation Therapy. Wolters Kluwer.

Chapter 8 — Treatments

What you'll learn

- The main external-beam techniques (3D-CRT, IMRT, VMAT, SBRT/SRS, TBI, electrons) and when each is used.
- How proton therapy uses the Bragg peak to spare tissue beyond the target.
- The three brachytherapy delivery types, their isotopes, applicators, and dose-specification points.
- How image guidance and record-and-verify systems confirm you are treating the right place, the right way, every day.
- The basics of how a linear accelerator makes its beam and how a daily treatment unfolds.
- Common acute toxicities, how imaging frequency trades off against margin, and how toxicity is graded with CTCAE version 5.

Why this matters. Treatment delivery is where the plan meets the patient. As a therapist, you are the last safety check before the beam turns on, and you manage patients through weeks of side effects. Knowing the why behind each technique helps you set up accurately, recognize when something is off, and explain care to anxious patients.

External-beam techniques

External-beam radiation therapy (EBRT) aims radiation at the patient from a machine outside the body, usually a **linear accelerator (linac)**. Over the decades the field has moved from simple boxes of radiation toward beams that wrap tightly around the tumor. Think of it as the difference between spray-painting a wall and using a fine airbrush.

3D conformal radiotherapy (3D-CRT)

3D-CRT uses a CT scan to build a three-dimensional picture of the tumor and nearby organs, then shapes each beam to match the tumor's outline from that beam's point of view (the "beam's-eye view"). The shaping is done with a **multileaf collimator (MLC)** — a bank of thin metal "leaves" that slide in and out to form a custom aperture, like a stencil made of movable tongue depressors.

In 3D-CRT each beam is uniform in intensity. It conforms the shape of the dose but cannot carve out concave notches around critical organs.

Intensity-modulated radiotherapy (IMRT)

IMRT goes one step further: it varies the intensity across each beam, not just the outline. By making some parts of a beam stronger and others weaker, IMRT can paint a dose that curves around sensitive structures — for example, wrapping around the spinal cord while still covering a tumor that hugs it.

IMRT is delivered two ways:

- **Step-and-shoot (segmental):** the MLC forms one shape, the beam turns on, then off; the leaves move to the next shape and the beam fires again. The dose is built from a stack of static "snapshots."
- **Sliding-window (dynamic):** the leaves move continuously while the beam stays on, sweeping across the field like a window shade. This is usually faster but demands tighter machine quality assurance.

> **Common mix-up:** IMRT vs VMAT. Both modulate intensity. The difference is delivery: traditional IMRT uses a set of fixed gantry angles (the gantry stops at each angle to treat), while VMAT delivers the dose as the gantry rotates continuously around the patient.

Volumetric-modulated arc therapy (VMAT)

VMAT is a form of IMRT delivered during one or more continuous gantry arcs. As the machine sweeps around the patient, three things change at once: the MLC leaves move, the gantry rotation speed varies, and the dose rate (how fast radiation comes out) varies. Because everything happens in a single rotation, VMAT is often much faster than fixed-angle IMRT — a treatment that took 10–15 minutes might finish in 2–3. Less time on the table means less chance for the patient to move.

Stereotactic radiosurgery and SBRT

Stereotactic means using precise coordinates to pinpoint a target. **Stereotactic radiosurgery (SRS)** treats targets in the brain; **stereotactic body radiation therapy (SBRT)**, also called SABR, treats targets in the body (lung, liver, spine).

These techniques share a recipe: a **few fractions** (often 1–5 treatments) delivering a **high dose per fraction**, aimed with **rigid immobilization** and image guidance for roughly **1–2 mm precision** [1]. Because the dose per session is so large, there is little room for error — the margins are tight and the falloff outside the target must be steep. AAPM Task Group 101 lays out the technical and safety requirements for SBRT [1].

> **Key Point:** A fraction is a single treatment session. Conventional courses use many small fractions (about 1.8–2 Gy each, sometimes 30–40 of them). SBRT/SRS flips this: very few fractions, each one large. More dose per visit means immobilization and imaging must be near-perfect.

Total body irradiation (TBI)

TBI irradiates the whole body, most often as part of **conditioning** before a bone-marrow or stem-cell transplant. The goal is to wipe out diseased marrow and suppress the immune system so the donor cells can engraft. A **commonly used** scheme is about **12 Gy in roughly 6 fractions** (for example, 2 Gy twice daily over three days), though protocols vary widely.

Because the lungs are sensitive to radiation, **lung shielding** (blocks that reduce dose to the lungs) is a standard part of many TBI techniques to lower the risk of radiation pneumonitis. The ILROG (International Lymphoma Radiation Oncology Group) guidelines describe contemporary TBI practice [6].

Electron beams

So far we have discussed **photon** (x-ray) beams, which penetrate deeply. **Electron beams** are different: electrons deposit their energy quickly and then stop, making them ideal for **superficial targets** — skin lesions, scars, or the chest wall after mastectomy.

A defining feature of electrons is that the dose is **normalized at d-max**, the depth of maximum dose. Beyond d-max the dose falls off rapidly, which protects deeper tissue. You choose the electron energy (in MeV) based on how deep the target sits: higher energy reaches deeper. A rough clinical rule of thumb is that the practical range in centimeters is about the energy in MeV divided by three.

Technique	Beam type	Intensity modulated?	Gantry motion	Typical fractionation	Best suited for
3D-CRT	Photon	No (shaped only)	Fixed angles	Conventional (many fractions)	Larger or simpler targets
IMRT	Photon	Yes	Fixed angles (step-and-shoot or sliding-window)	Conventional	Targets near critical organs
VMAT	Photon	Yes	Continuous arc(s)	Conventional	Same as IMRT, but faster delivery
SBRT / SRS	Photon	Yes (often)	Arcs or many beams	Few fractions, high dose each	Small, well-defined targets with rigid setup

Electrons and TBI are specialized techniques rather than direct alternatives to the photon methods above, so they sit outside this comparison.

Particle therapy: protons

Proton therapy uses charged particles instead of x-rays, and its key advantage is a physics phenomenon called the **Bragg peak**.

When a proton beam enters tissue, it deposits relatively little dose along the way. Then, at a depth determined by its energy, it dumps most of its energy in a sharp spike — the Bragg peak — and stops. Almost no dose continues past that point. Picture a car that coasts gently down a street and then slams on the brakes at one exact house, with nothing happening beyond it.

A single Bragg peak is very narrow, so to cover a whole tumor the team stacks many peaks of different energies side by side. The result is a **spread-out Bragg peak (SOBP)** — a plateau of uniform dose across the target depth, with **minimal exit dose** beyond it. This sparing of tissue *behind* the tumor is why protons are attractive for children and for tumors near critical structures.

> **Key Point:** The headline benefit of protons is not a higher tumor dose — it is the near-absence of exit dose past the target, thanks to the Bragg peak.

Brachytherapy

Brachytherapy (from the Greek *brachy*, "short") places the radioactive source very close to — or inside — the tumor. Because dose falls off steeply with distance, brachytherapy delivers a high dose to the target while sparing nearby tissue.

Sources are classed by how fast they deliver dose: **dose rate**.

Type	Dose rate	Isotope	Typical use
HDR	High-dose-rate	**Iridium-192**	Stepping-source treatments (gynecologic, breast, etc.)
LDR intracavitary	Low-dose-rate	**Cesium-137**	Classic GYN intracavitary inserts
Permanent seeds	Low (permanent)	**Iodine-125**	Prostate seed implants left in place

> **Common mix-up:** HDR vs LDR — and the isotopes. **HDR uses Iridium-192**, a single tiny source on a wire that *steps* through the applicator under computer control, pausing at programmed positions (dwell points) for set times. **LDR intracavitary uses Cesium-137** and delivers dose slowly over hours to days. **Permanent prostate seed implants use Iodine-125**, left in the gland permanently to decay away. Keep these three straight — swapping the isotopes is a classic exam trap.

Applicators

The source is delivered through an **applicator** matched to the anatomy:

- **Tandem and ovoids (Fletcher-Suit):** a central tube (tandem) placed through the cervical canal into the uterus, plus two side capsules (ovoids) in the vaginal fornices. This is the workhorse for **gynecologic intracavitary** treatment of cervical cancer.
- **Interstitial needles/catheters:** thin needles or catheters placed directly through the tissue/tumor (for example, in the breast or prostate) so the source can pass through them.
- **Intraluminal catheters:** a catheter placed inside a body **lumen** (a hollow passage such as the esophagus or a bronchus) so the source can treat the lining from within.

Dose specification points

Gynecologic brachytherapy historically reports dose at two anatomic landmarks:

- **Point A:** roughly 2 cm up from the cervical os and 2 cm lateral to the tandem — near where the uterine artery crosses the ureter. It approximates the dose to the **target/ paracervical tissue**.
- **Point B:** 3 cm lateral to Point A (5 cm from midline). It approximates dose at the **pelvic sidewall**, near the lymph nodes.

ICRU Report 38 standardized this older point-and-volume reporting for intracavitary GYN brachytherapy, and **ICRU Report 89** updated it for modern image-based (CT/MRI) planning that reports dose to target volumes and organs at risk rather than relying on points alone [4][5].

Image guidance and verification at delivery

Planning a beautiful dose distribution is useless if the patient is not in the right position. **Image-guided radiation therapy (IGRT)** confirms the target's location at the machine, on the day, before the beam goes on.

- **Portal imaging:** an x-ray taken with the treatment beam (or a low-dose kV beam) to check bony anatomy against the plan. Modern systems use an **electronic portal imaging device (EPID)**, a flat-panel detector.
- **Cone-beam CT (CBCT):** the gantry rotates while a kV x-ray source and detector acquire a 3D volume, letting you align soft tissue and bone in three planes — not just a flat shadow.
- **Fiducial or soft-tissue matching: fiducials** are small markers (often gold seeds) implanted in or near the tumor; they show up clearly on imaging so you can align to the tumor itself. When good soft-tissue contrast exists (as on CBCT), you may match directly to the organ.

After imaging, the therapist (or therapist and physician, per policy) reviews the match and applies couch shifts to correct any offset before treating.

Record-and-verify systems

A **record-and-verify (R&V)** system is the safety brain of the treatment room. It holds the approved plan parameters — energy, MLC shapes, gantry and collimator angles, monitor units, couch position — and **compares planned versus delivered** values in real time. If anything falls **outside tolerance**, the system **interlocks** (stops the beam) rather than treating the wrong way. It also records every fraction, building the treatment history.

> **Key Point:** IGRT answers "is the patient in the right place?" R&V answers "is the machine doing exactly what the plan said?" You need both.

Machine operation basics

How the linac makes the beam

A linear accelerator builds its beam in stages:

1. An **electron gun** injects electrons into an **accelerating waveguide**.
2. Microwaves (from a magnetron or klystron) accelerate the electrons to nearly the speed of light down the waveguide.
3. For **electron** treatments, that pencil of electrons is spread out (by scattering foils or scanning) and used directly.
4. For **photon (x-ray)** treatments, the electrons slam into a high-density **target**, producing x-rays by bremsstrahlung ("braking radiation").
5. A **flattening filter** (in conventional modes) evens out the beam; the **primary and secondary collimators** plus the **MLC** shape it; and **monitor chambers** measure the output in **monitor units (MU)**.

Beam modifiers

Several devices tailor the beam:

- **Wedges** tilt the dose gradient across the field (physical metal wedges or "virtual" wedges made with moving jaws).
- **Bolus** is tissue-equivalent material laid on the skin to pull dose up to the surface (useful for skin or scar treatments).
- **Blocks/MLC** shape the field and shield normal tissue.
- **Compensators** correct for sloping or irregular body surfaces.

Daily setup workflow

A typical fraction follows a predictable rhythm:

1. **Identify the patient** with at least two identifiers — this is non-negotiable.
2. **Position and immobilize** using the same devices used at simulation (masks, vac-bags, breast boards).
3. **Align** to skin marks/tattoos and room lasers to the planned isocenter.
4. **Image** (portal, CBCT, or fiducial match) and apply any couch shifts.
5. **Treat**, monitoring the patient by camera and audio, watching the R&V console for interlocks.
6. **Document** the fraction, image approvals, and any patient-reported issues.

On-treatment management

You will see patients several days a week for weeks, so you are often the first to notice side effects. **Acute toxicities** appear during or shortly after treatment (versus late effects that show up months to years later).

Common acute toxicities depend on what is in the beam:

- **Mucositis:** painful inflammation of the mouth/throat lining (head-and-neck treatment).
- **Esophagitis:** painful swallowing from an inflamed esophagus (chest treatment).
- **Dermatitis:** skin reaction ranging from redness to moist peeling, especially in skin folds and high-dose surface areas.
- **Myelosuppression:** drop in blood counts when large volumes of bone marrow are irradiated (broad pelvic/spine fields, TBI). Watch for low white cells (infection risk), low platelets (bleeding), and anemia (fatigue).

The imaging-versus-margin trade-off

Every plan includes a **PTV (planning target volume) margin** — extra room around the tumor to account for setup uncertainty and motion. There is a trade-off:

- **Image more often** (e.g., daily CBCT) → you catch and correct day-to-day shifts → you can use a **smaller margin** → less normal tissue irradiated, but more imaging dose and time.
- **Image less often** → you must use a **larger margin** to stay safe → more normal tissue in the high-dose region.

> **Key Point:** Frequent image guidance "buys back" margin. Tighter, more accurate setup lets the planner shrink the PTV and spare healthy tissue — one big reason IGRT became standard.

Supportive care

Good supportive care keeps patients on schedule:

- **Skin:** gentle washing, prescribed moisturizers, avoid friction and sun; topical or dressing care for moist desquamation.
- **Mucositis/esophagitis:** topical/systemic analgesics, "magic mouthwash"-type rinses, soft diet, nutrition support, hydration.
- **Nutrition and weight:** monitor closely in head-and-neck and GI patients; dietitian referral.
- **Monitoring counts:** regular CBCs during marrow-heavy treatment.

Grading toxicity with CTCAE v5

Clinicians grade side effects with the **Common Terminology Criteria for Adverse Events, version 5 (CTCAE v5)** [7]. It assigns each toxicity a severity grade so the whole team speaks the same language:

Grade	General meaning
1	Mild; usually no intervention needed
2	Moderate; minimal/local intervention; affects some daily activities
3	Severe but not immediately life-threatening; may need hospitalization
4	Life-threatening; urgent intervention
5	Death related to the toxicity

Common mix-up: CTCAE grade describes **severity**, not how soon a toxicity appears. A Grade 3 acute dermatitis and a Grade 3 late fibrosis are both "severe," but one is early and one is late — timing (acute vs late) is a separate idea from grade.

Check yourself

1. **A patient is scheduled for a single high-dose treatment to a small brain metastasis with a rigid frame. What technique is this, and roughly what precision is expected?** Stereotactic radiosurgery (SRS): one (or very few) high-dose fractions with rigid immobilization and roughly 1–2 mm precision [1].
2. **What is the practical difference between IMRT and VMAT?** Both modulate beam intensity. Traditional IMRT treats from fixed gantry angles; VMAT delivers the dose during one or more continuous gantry arcs, usually faster.

3. **Match each brachytherapy type to its isotope: HDR, LDR intracavitary, permanent prostate seeds.** HDR uses Iridium-192; LDR intracavitary uses Cesium-137; permanent prostate seeds use Iodine-125.
4. **Why do proton beams spare tissue beyond the tumor better than x-rays?** Protons deposit most of their energy at the Bragg peak and then stop, leaving minimal exit dose; stacking peaks of different energies makes a spread-out Bragg peak (SOBP) covering the target.
5. **A department switches from weekly portal imaging to daily CBCT. How might this affect the PTV margin, and why?** Daily imaging catches and corrects setup variation, so the planner can use a smaller PTV margin, sparing more normal tissue (at the cost of more imaging time and dose).
6. **What does a record-and-verify system do when a delivered parameter falls outside tolerance?** It interlocks and stops the beam, preventing delivery that does not match the approved plan, and it logs the event.

Chapter references

Free, full-text sources

1. IAEA — Radiation Oncology Physics: A Handbook for Teachers and Students (free PDF; external-beam techniques, brachytherapy, electrons): https://www-pub.iaea.org/MTCD/Publications/PDF/Pub1196_web.pdf.
2. AAPM Reports — stereotactic body radiotherapy (TG-101) and related delivery/QA, free: https://www.aapm.org/pubs/reports/.
3. NCI — CTCAE v5.0 (Common Terminology Criteria for Adverse Events), free: https://ctep.cancer.gov/protocoldevelopment/electronic_applications/ctc.htm.
4. NCI — PDQ treatment summaries (modality selection by site): https://www.cancer.gov/types.

For deeper reading (library / standards)

1. ICRU Reports 38 and 89 — brachytherapy dose and reporting (ICRU; purchase).
2. Khan FM, Gibbons JP. The Physics of Radiation Therapy. Wolters Kluwer; Gunderson & Tepper, Clinical Radiation Oncology.

A Note on Sources & Accuracy

Every chapter in this book ends with its own numbered reference list, so you can trace any fact back to an authoritative source. Those sources fall into a few trusted families:

- **The standards bodies.** The ICRU (International Commission on Radiation Units and Measurements) defines the target volumes and how we report dose. The AAPM (American Association of Physicists in Medicine) publishes the "Task Group" reports that set quality-assurance and dose-calculation practice (TG-51, TG-66, TG-71, TG-76, TG-101, TG-142, TG-179, and more). The NCRP and the U.S. NRC (Nuclear Regulatory Commission, 10 CFR 20) set radiation-protection rules and dose limits. The AJCC defines cancer staging, and the NCCN publishes treatment guidelines.
- **The professional organizations.** The ASRT (American Society of Radiologic Technologists) publishes the practice standards and scope of practice for radiation therapists, and the ARRT publishes the Standards of Ethics.
- **The textbooks.** Three classics underpin most of this material: Khan's The Physics of Radiation Therapy, Washington & Leaver's Principles and Practice of Radiation Therapy, and Gunderson & Tepper's Clinical Radiation Oncology. For radiobiology, Hall & Giaccia's Radiobiology for the Radiologist is the standard. These are commercial books — find them through a library or program.

Wherever a fact can be checked in a **free, full-text** source, the chapter references now lead with it, so you can click straight through and verify for yourself. The most useful free resources are the IAEA's Radiation Oncology Physics: A Handbook for Teachers and Students (a complete physics text, free PDF), the AAPM Task Group reports, the U.S. NRC's 10 CFR 20, the CDC's radiation-safety pages, and the NCI's PDQ cancer summaries. The commercial textbooks above are listed under "for deeper reading."

A few accuracy notes worth carrying with you, because they are exactly the places where casual study materials often go wrong:

- **Dose limits in this book are the U.S. NRC values** (occupational 50 mSv/year, public 1 mSv/year, declared-pregnant worker 5 mSv over the gestation). International (ICRP) recommendations differ — for the U.S. registry, learn the NRC numbers.
- **Brachytherapy isotopes:** high-dose-rate (HDR) uses **Iridium-192**; low-dose-rate intracavitary uses **Cesium-137**; permanent prostate seeds use **Iodine-125**. Keep them straight.
- **The inverse-square law** is $D_2 = D_1 \times (d_1/d_2)^2$, where the d's are distances.
- **Doses and fractionation schemes are examples,** not universal truths. Real prescriptions follow the treating physician's judgment and current protocols.

Key Point: When a study aid and a standard disagree, the standard wins. This book points you to the standards on purpose — so you are always one click away from the source of truth.

Keep Studying

You do not have to learn everything at once. A simple, repeatable rhythm beats a heroic all-nighter every time:

1. **Read one chapter** for understanding. Do not highlight everything — just follow the thread.
2. **Re-read the Key Point and Common mix-up boxes.** These are your high-yield review.
3. **Do the Check-yourself questions** without peeking. Getting one wrong is useful — it shows you exactly where to look again.
4. **Come back the next day** and re-answer yesterday's questions before starting the new chapter. This "spaced" review is what moves knowledge into long-term memory.
5. **Mix it up.** Once you have read a few chapters, jump around. The exam will not hand you topics in order, so practice retrieving them out of order too.

When the eight chapters feel comfortable, switch to practice questions and a timed run-through, so the clock and the multiple-choice format feel familiar before exam day.

You are studying to do something genuinely important: to set up and deliver radiation safely and accurately for people on one of the hardest days of their lives. Every concept in this book is in service of that. Be patient with yourself, trust the steady rhythm, and keep going.
